Computer Vision —— Stereo Vision

2026/10/27

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Slides from CV course with R. Fergus, Jana Kosecka, A. Zisserman & S. Lazebnik, Antonio Torralba

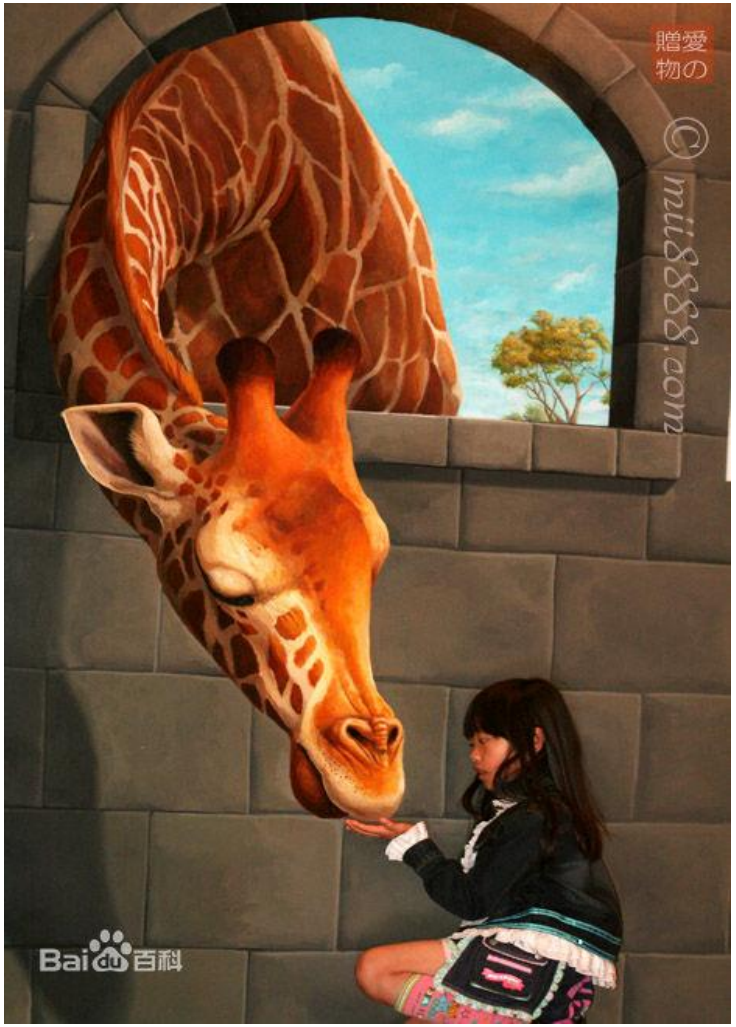
Agenda

- Depth perception
- Stereo vision
- Stereo Matching
- 3D Construction

Depth Perception

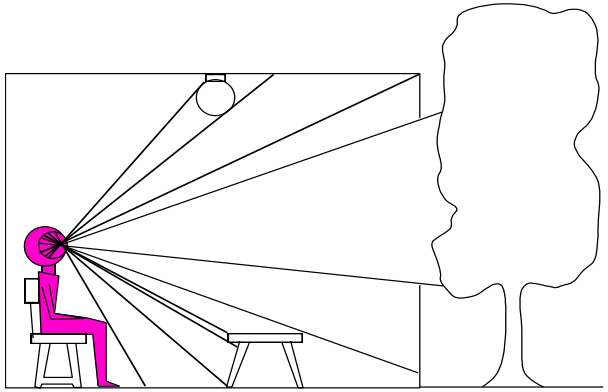
What is depth perception

Monocular depth perception

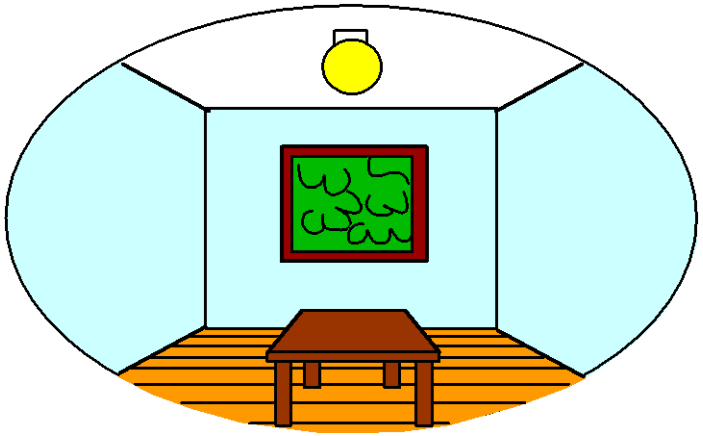


Depth Perception

2D images project 3D points into 2D:



Point of observation

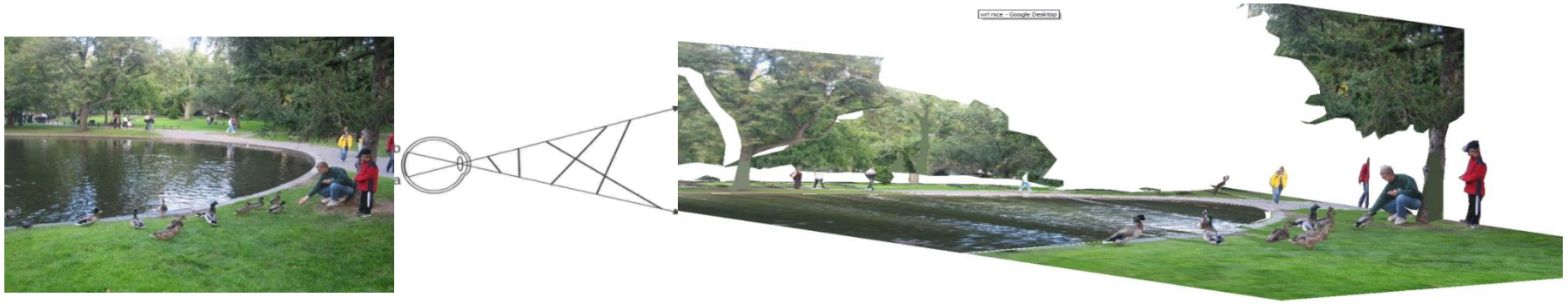


3D world

2D image

Figures © Stephen E. Palmer, 2002

Depth Perception



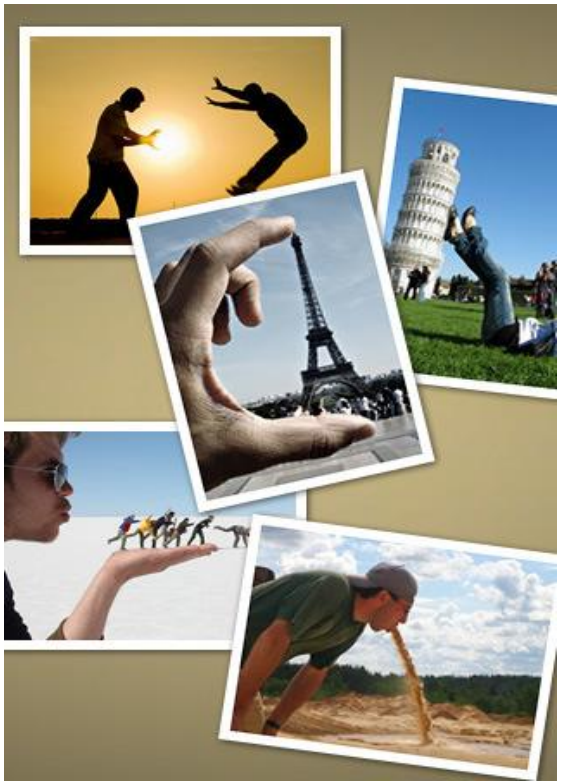
The inverse problem

Depth Perception

Applications

- 3D movies
- photography





Monocular cues to depth

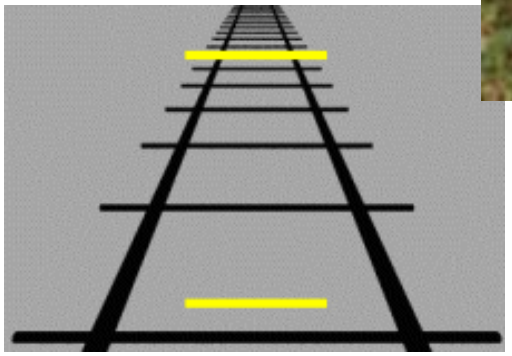
- **Absolute depth cues**: (assuming known camera parameters) these cues provide information about the absolute depth between the observer and elements of the scene
- **Relative depth cues**: provide relative information about depth between elements in the scene (this point is twice as far at that point, …)

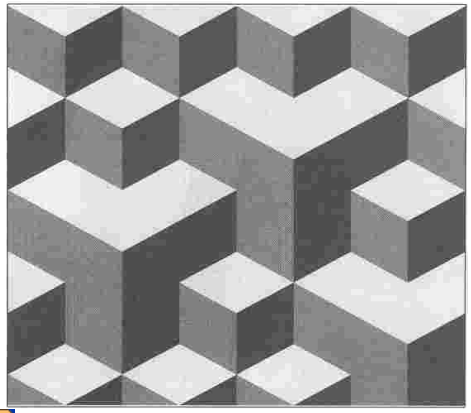
Monocular depth perception













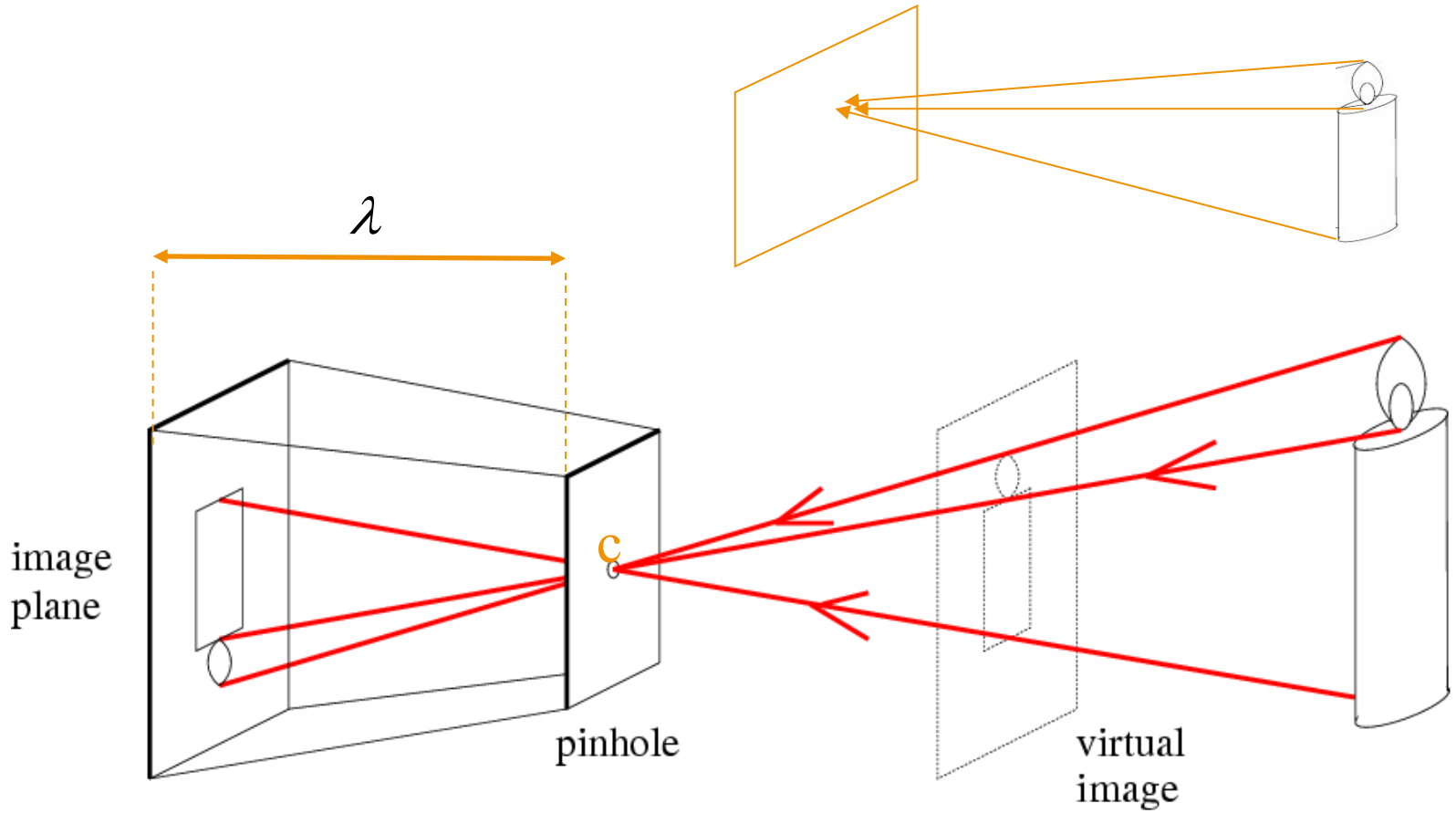
Monocular depth perception







Depth Perception

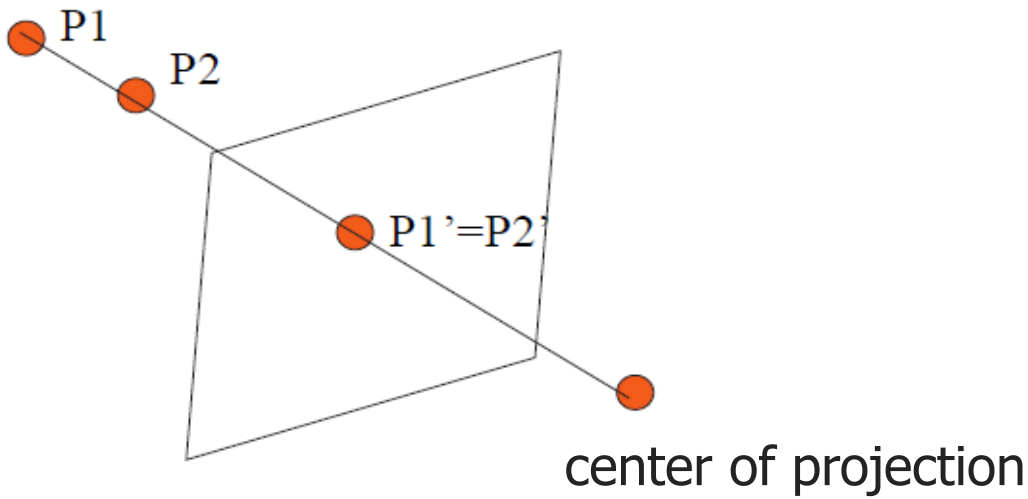


Pinhole camera

λ = focal distance， c = center of camera

Forsyth & Ponce

Depth Perception

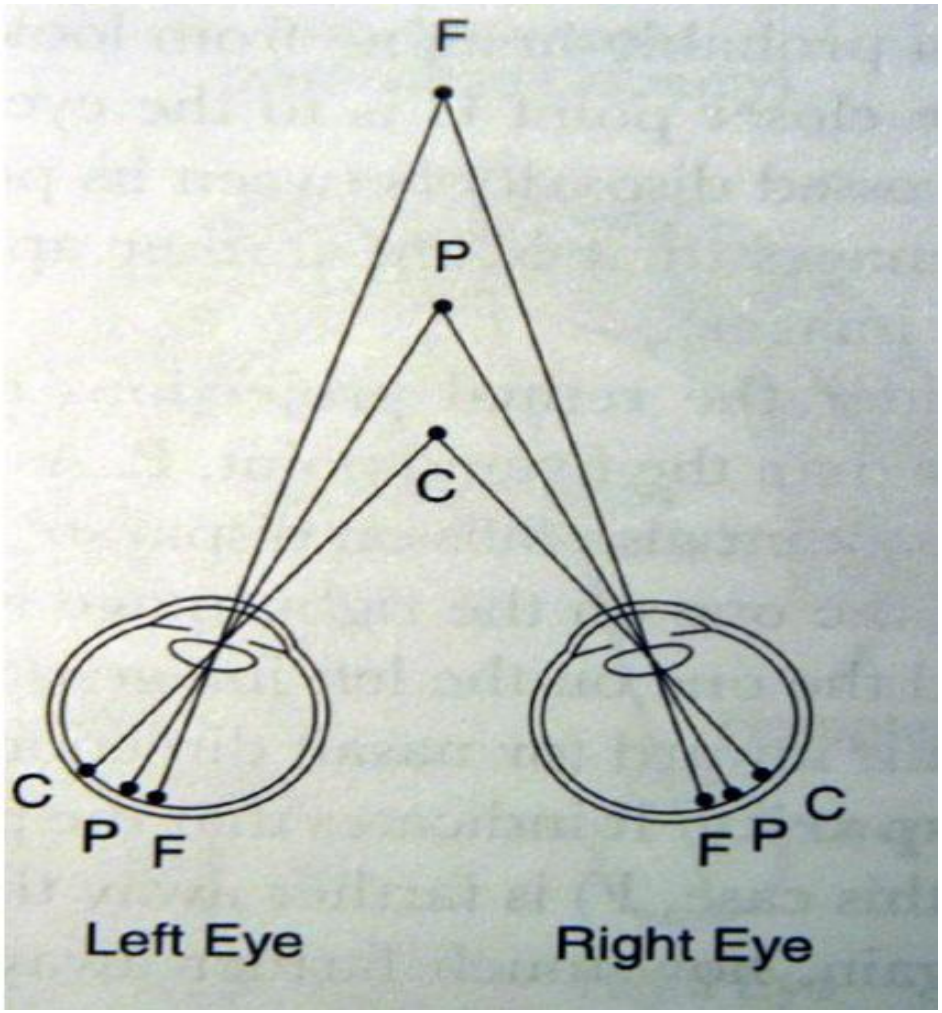


- 3D points on the same viewing line have the same 2D image:
  - 2D imaging results in depth information loss

Agenda

- Depth perception
- Stereo vision
- Stereo Matching
- 3D Construction

Stereo Vision



Stereo Vision

Applications：

Path planning （robots、automatic drive）

3D Reconstruction

Virtual reality

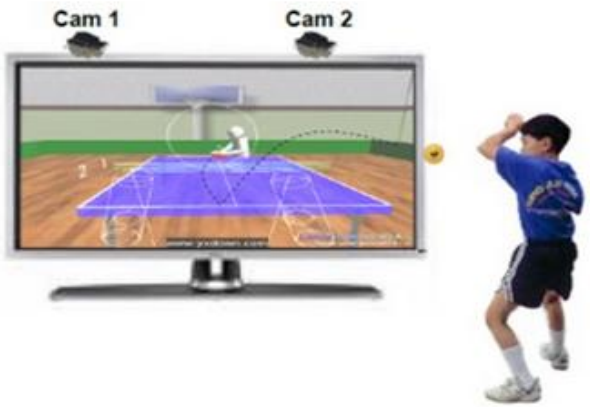
Non-contact measurement

Street view



Stereo Vision

Applications：





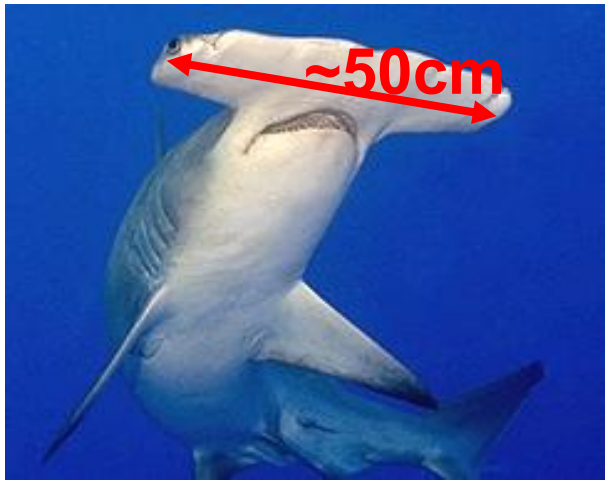




Stereo Vision

Obtain depth information from multiple images

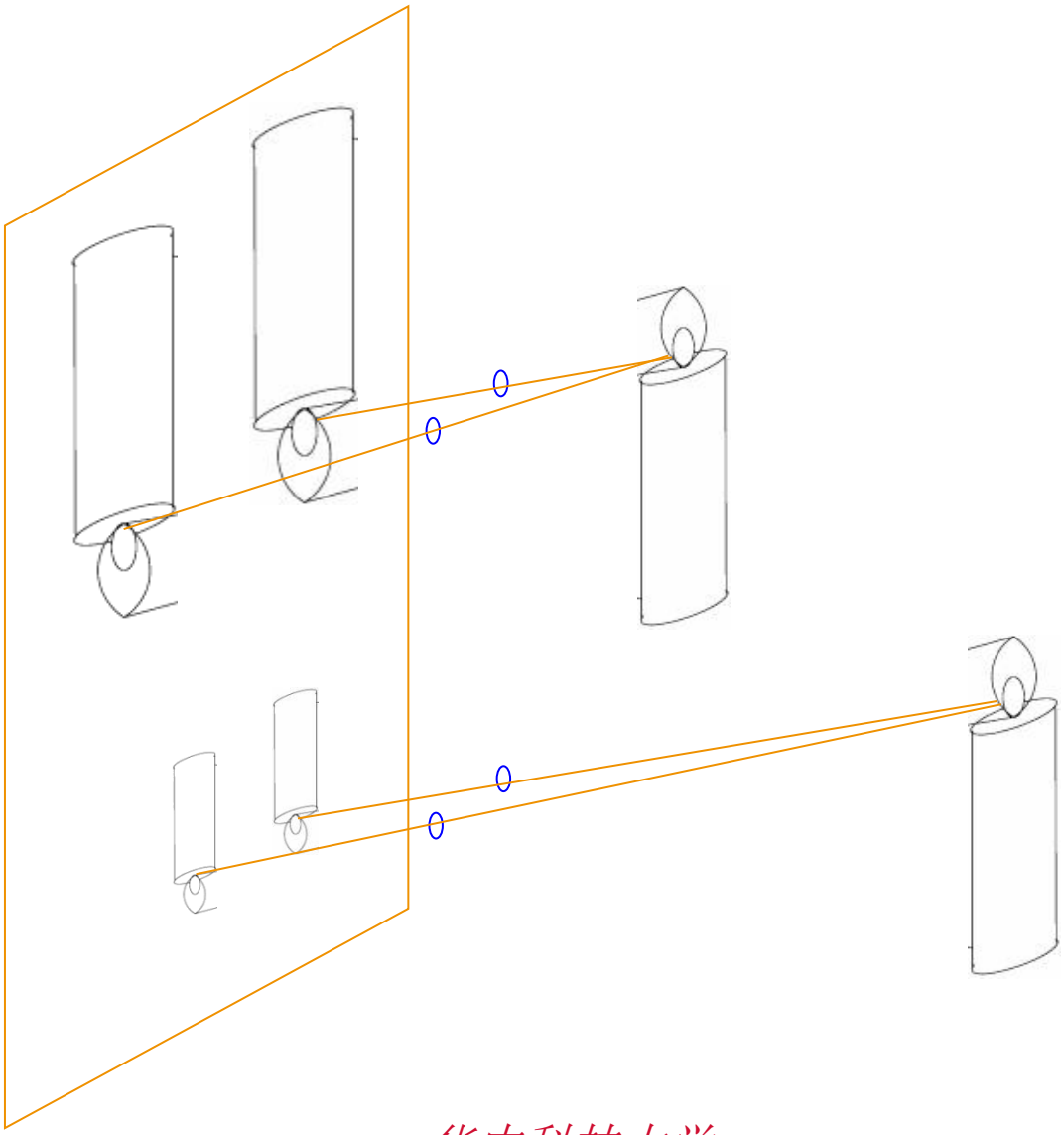






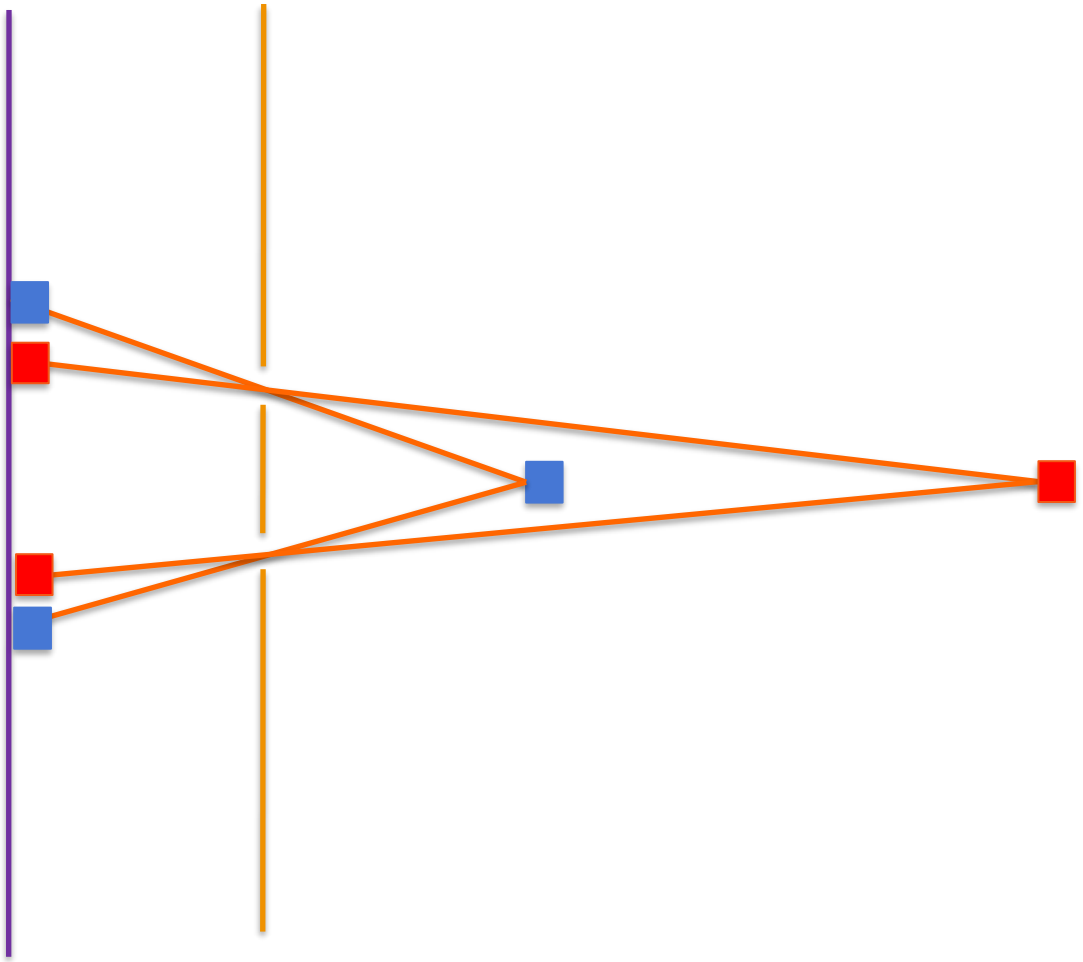


Stereo Vision



Stereo Vision

- two pinholes



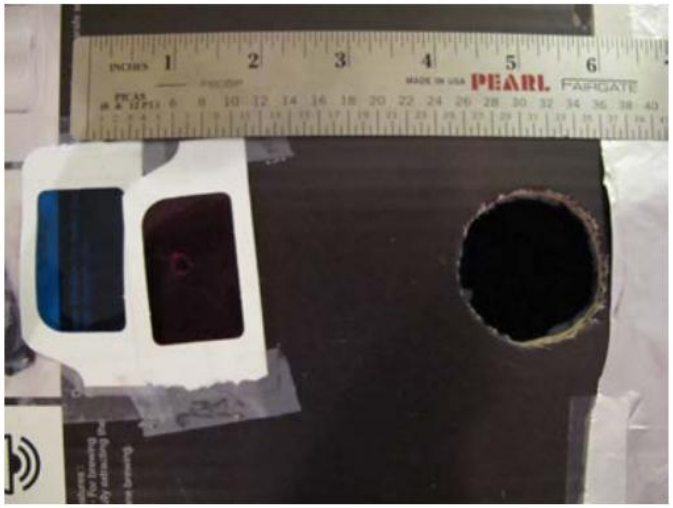
Stereo Vision

- 3D Anaglyph red-green



Anaglyph pinhole camera





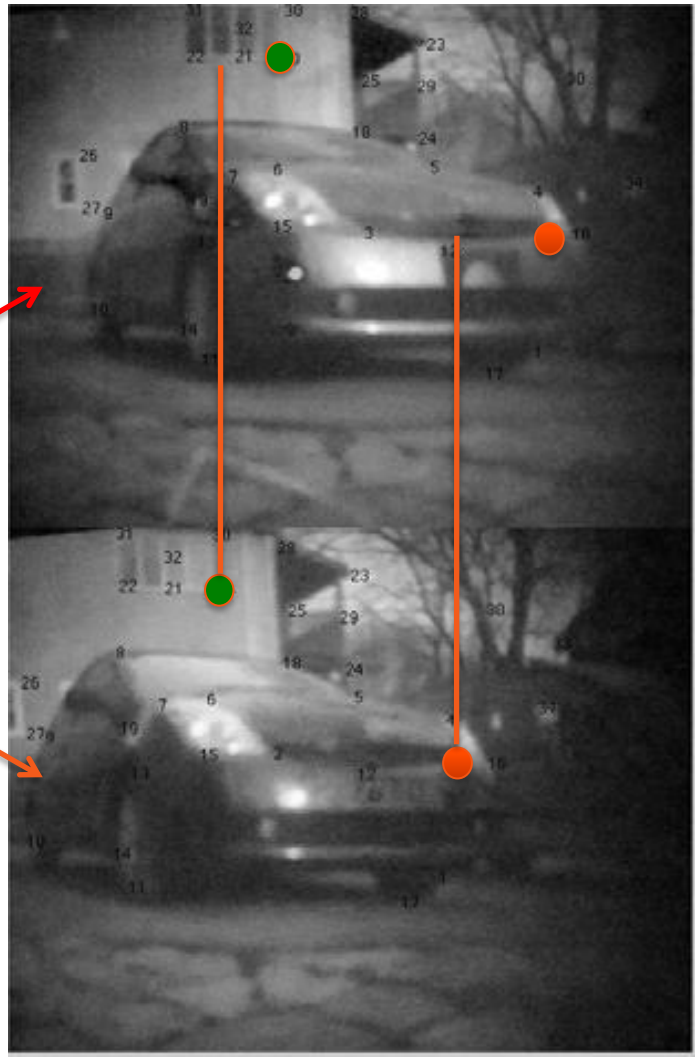




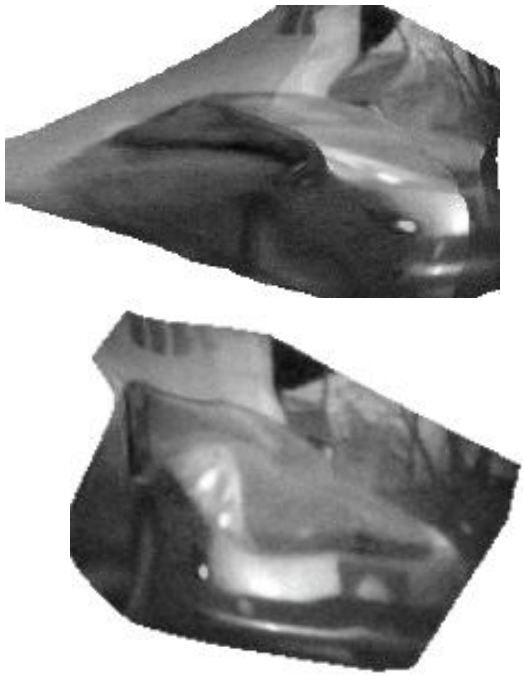
Stereo Vision

Anaglyph image





New view

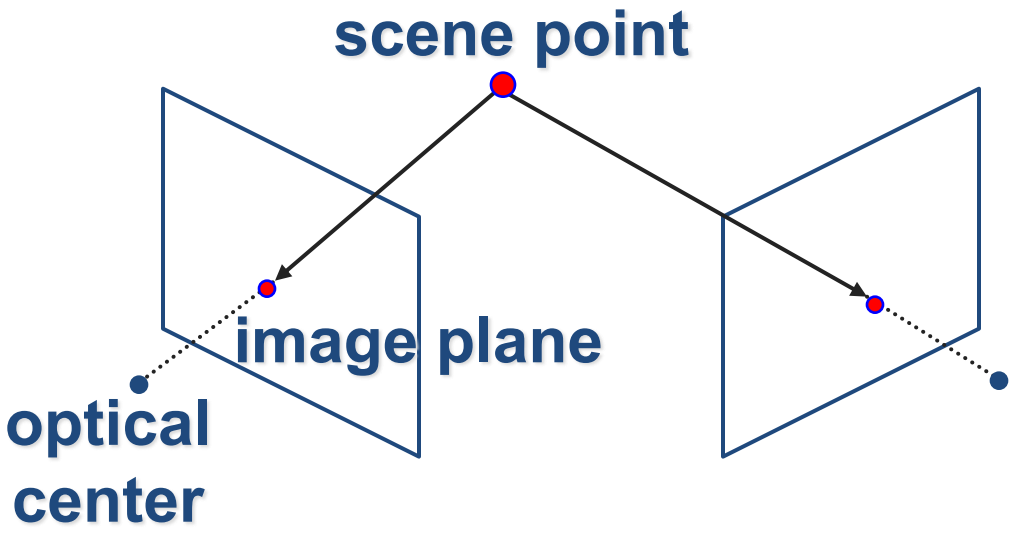


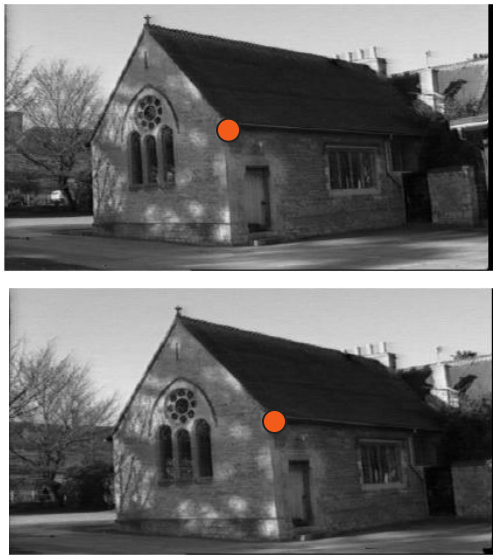
Stereo Vision

1. Camera calibration
2. Images capture
3. Feature extraction
4. Stereo matching
5. 3d information construction
6. Post processing

Estimating depth with stereo

- **Stereo**: shape from disparities between two views
- We'll need to consider:
  - Info on camera pose ("calibration")
  - Image point correspondences

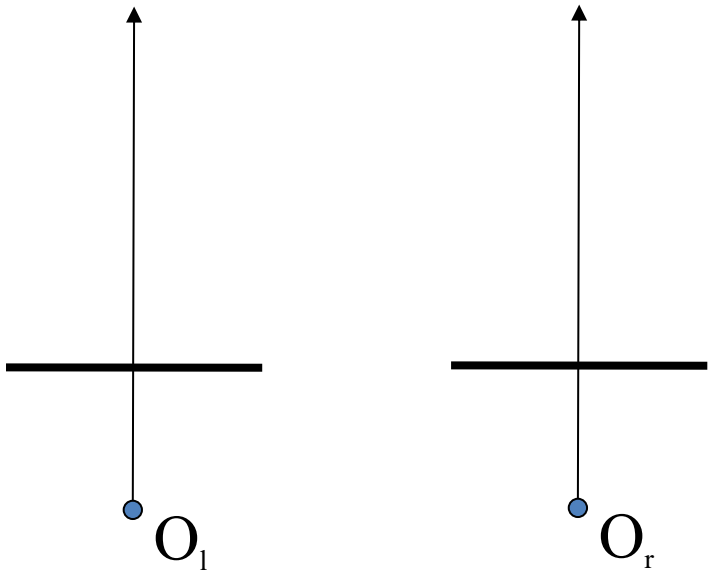


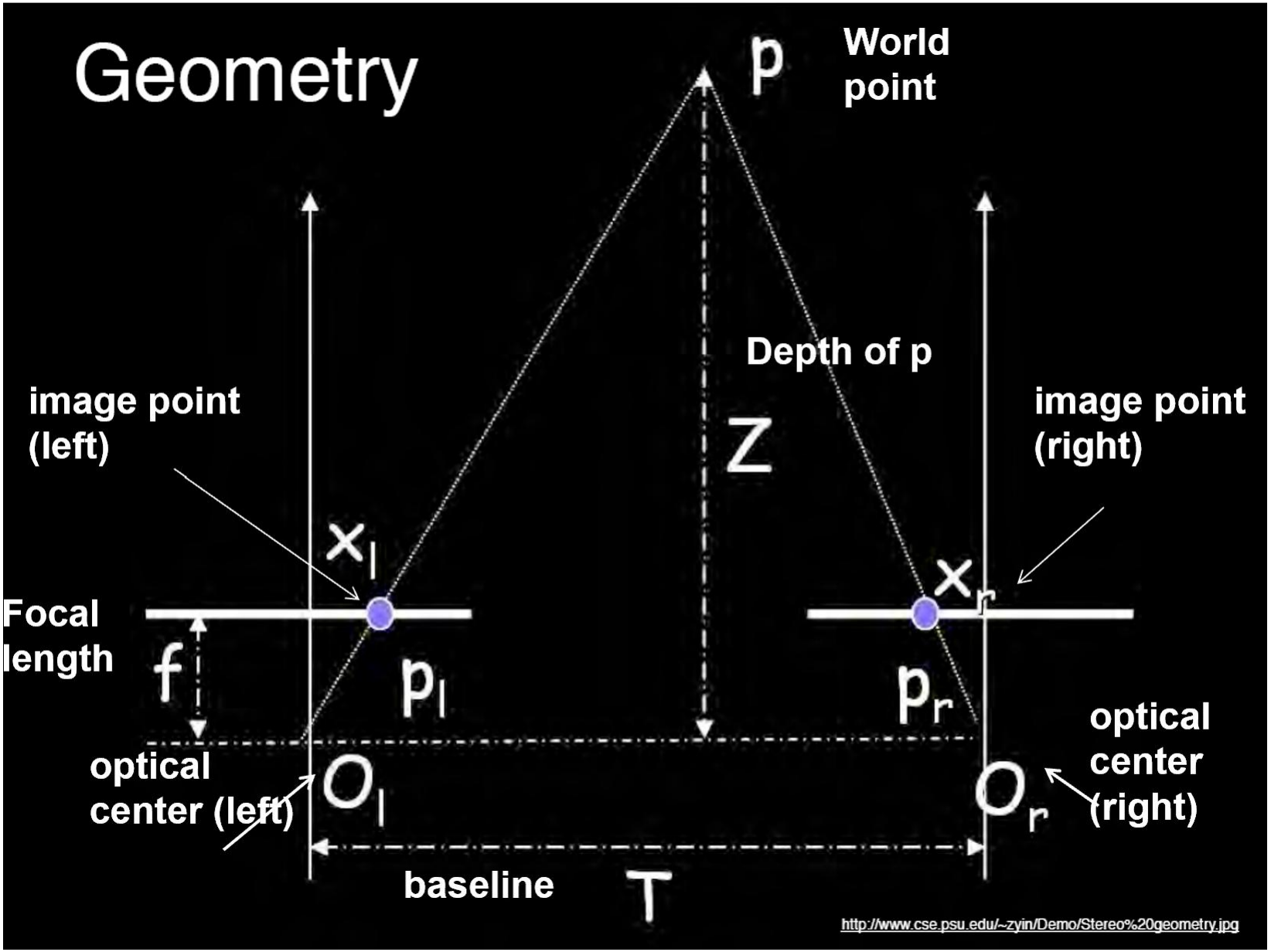


Slide credit: Kristen Grauman

Geometry for a simple stereo system

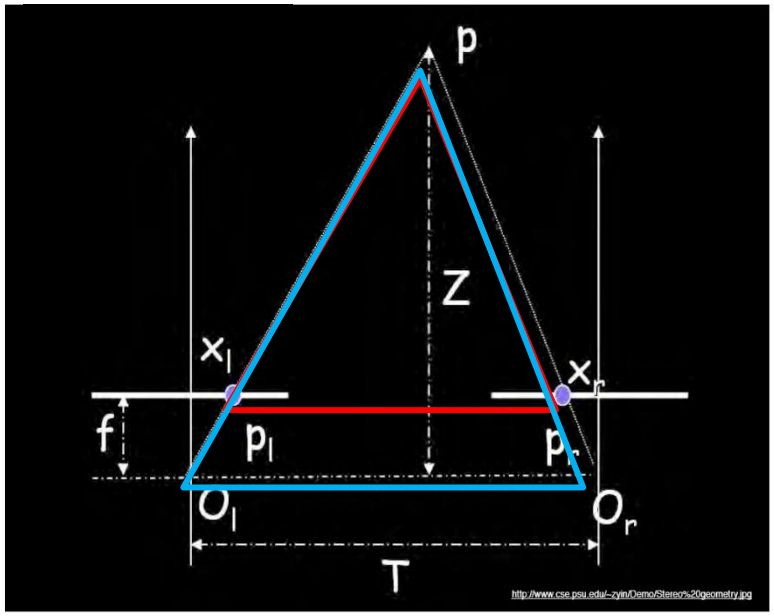
- Assume a simple setting:
  - Two identical cameras
  - parallel optical axes
  - known camera parameters (i.e., calibrated cameras).




Geometry
p
World point
Depth of p
Z
image point (left)
image point (right)
x_l
x_r
Focal length
f
p_l
p_r
optical center (left)
O_l
O_r
optical center (right)
baseline
T
http://www.cse.psu.edu/~zyin/Demo/Stereo%20geometry.jpg

Geometry for a simple stereo system

- Assume parallel optical axes, known camera parameters (i.e., calibrated cameras). We can triangulate via:

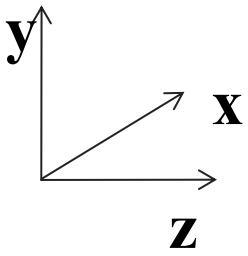


Similar triangles (p_l, P, p_r) and (O_l, P, O_r):

$$\frac{T + x_l - x_r}{Z - f} = \frac{T}{Z}$$

$$Z = f\frac{T}{x_r - x_l}$$

disparity

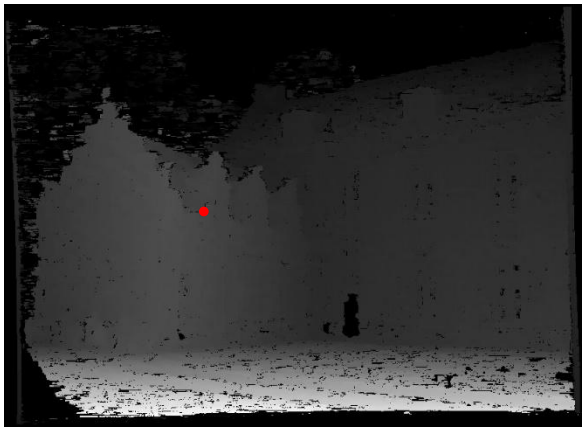


Slide credit: Kristen Grauman

Depth from disparity

image I(x,y) **Disparity map D(x,y)** **image I´(x´,y´)**







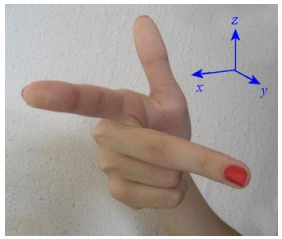
(x´,y´)=(x+D(x,y), y)

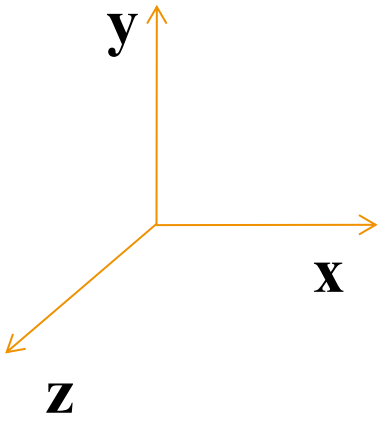
$$Z = f\frac{T}{d}$$

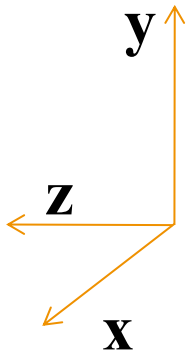
Slide credit: Kristen Grauman

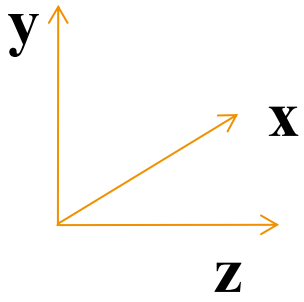
Depth from disparity

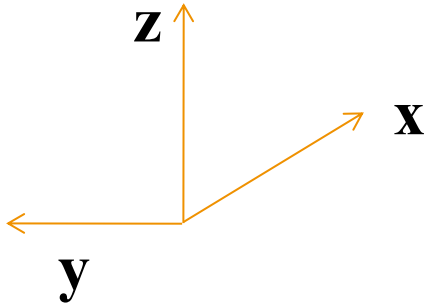
- Right - handed system





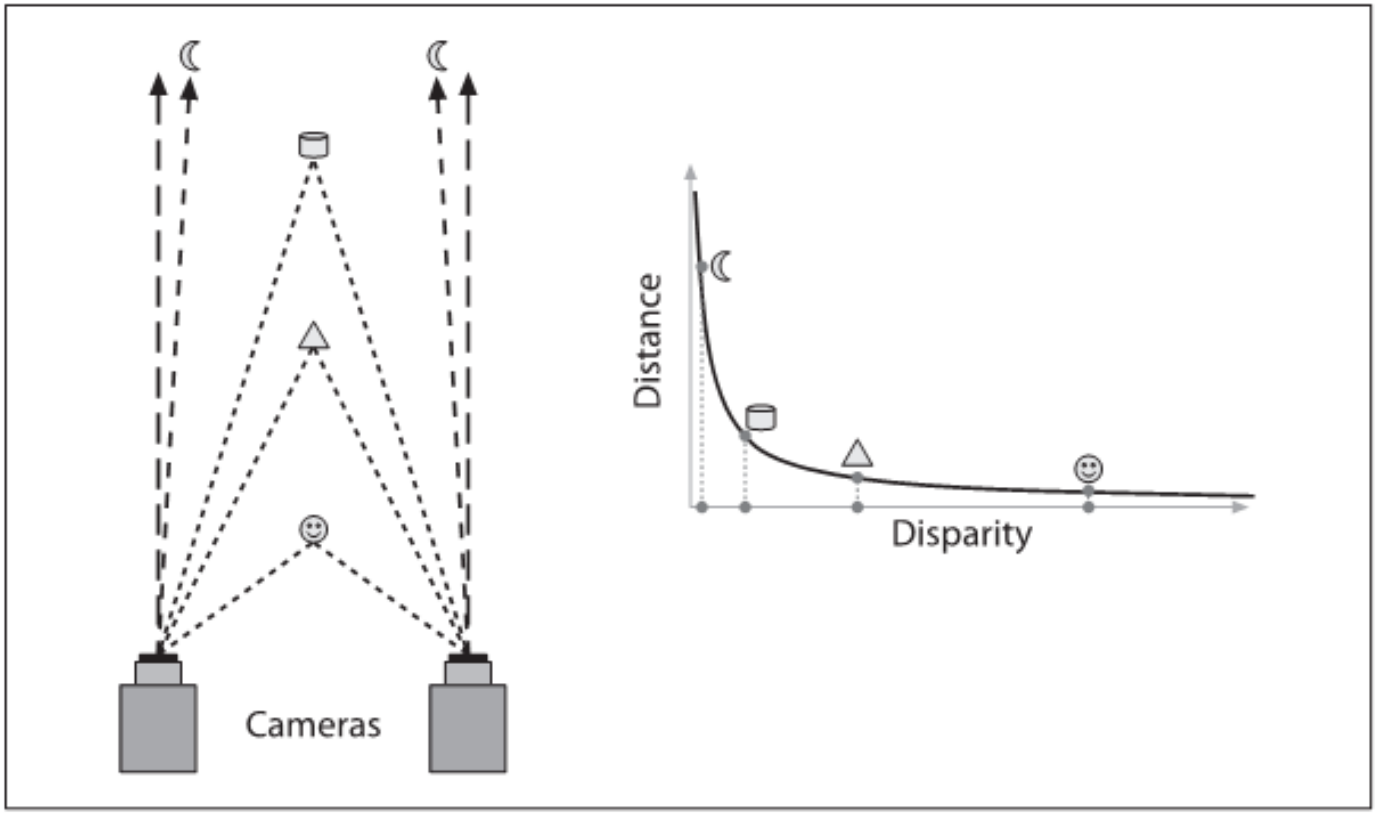






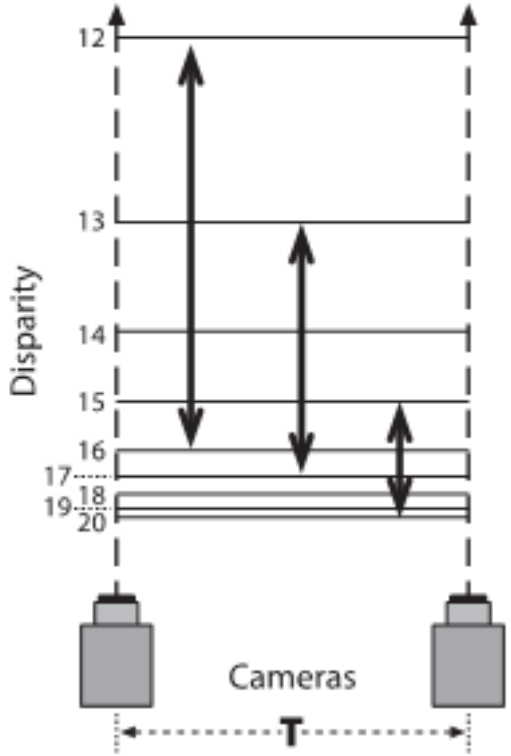
Depth from disparity

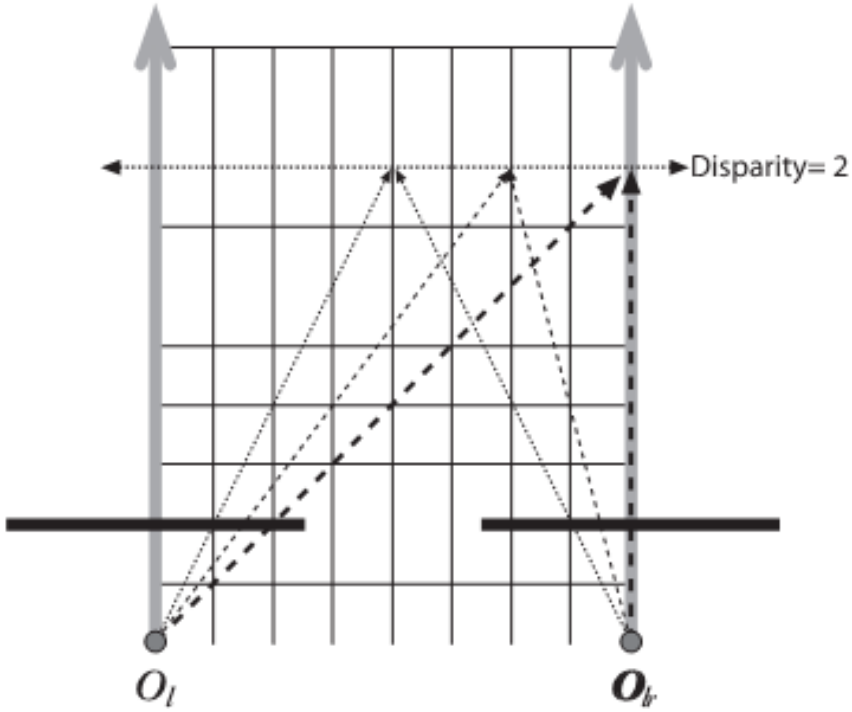
$$Z = f\frac{T}{d}$$



Depth from disparity

- Disparity and depth





Disparity and depth

1. Disparity and depth

$x_{1e} = x_1 + e$， $d_{1e} = x_1 + e - x_2 = d + e$

$$\Delta Z = Z_{1e} - Z = f\frac{B}{d_{1e}} - f\frac{B}{d} = \frac{-fBe}{d(d+e)}$$

$$\frac{fB}{Z-f} = x_1 - x_2 = d$$

$$\Delta Z = \frac{-e(Z-f)^2}{fB + e(Z-f)} \approx \frac{-eZ^2}{fB + eZ}$$

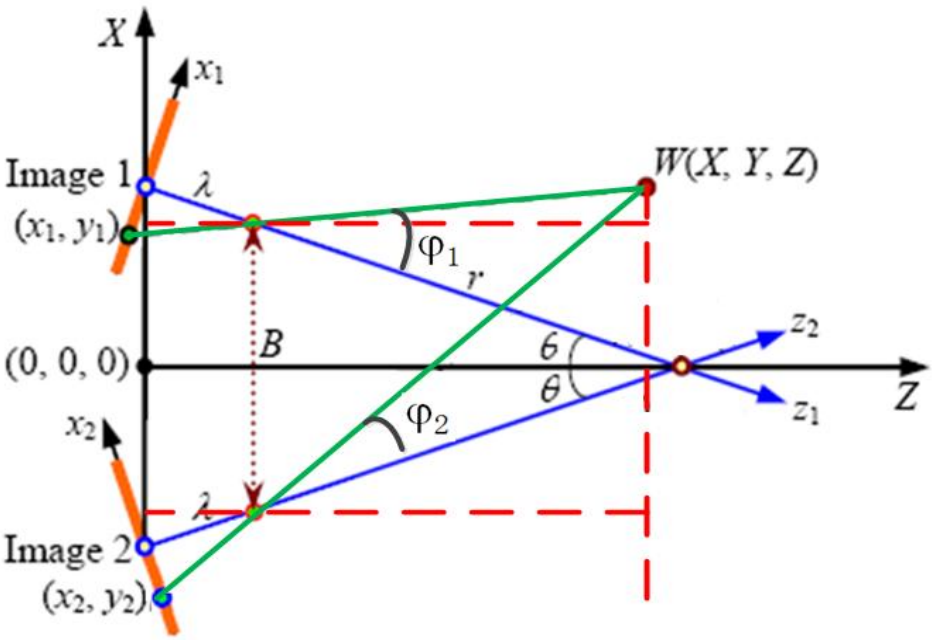
Stereo Vision

$$\frac{X-\frac{B}{2}}{(Z-\lambda\cos\theta)}=\tan(\phi_1-\theta)$$

$$\frac{X+\frac{B}{2}}{(Z-\lambda\cos\theta)}=\tan(\phi_2+\theta)$$

$$\phi_1 = arc\tan(\frac{x_1}{\lambda})$$

$$\phi_2 = arc\tan(\frac{-x_2}{\lambda})$$



$$Z=\frac{B}{\tan(\phi_2+\theta)-\tan(\phi_1-\theta)}+\lambda\cos\theta$$

Agenda

- Depth perception
- Stereo vision
- Stereo Matching
- 3D Construction

The objective

Given two images of a scene acquired by known cameras compute the 3D position of the scene (structure recovery)



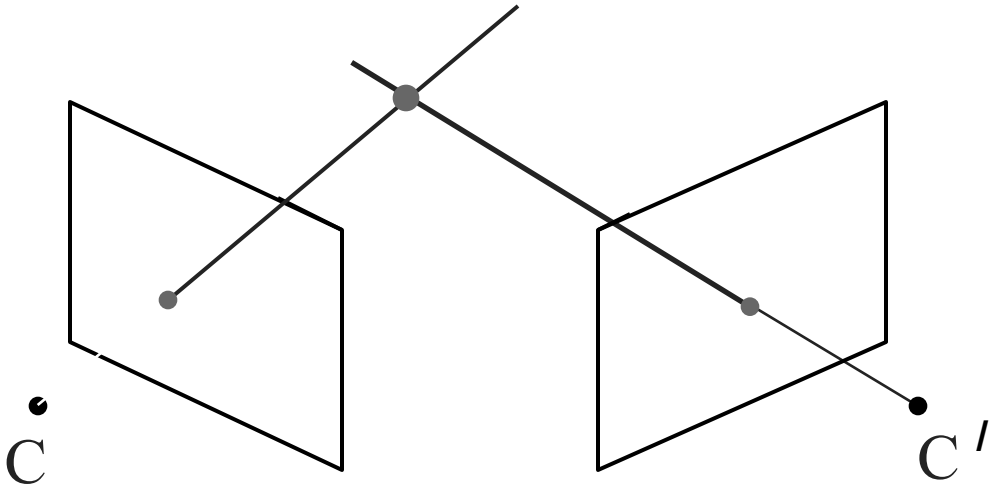
Basic principle: triangulate from corresponding image points

- Determine 3D point at intersection of two back-projected rays

Corresponding points are images of the same scene point



Triangulation



The back-projected points generate rays which intersect at the 3D scene point

An algorithm for stereo reconstruction

1. For each point in the first image determine the corresponding point in the second image

   (this is a search problem)

2. For each pair of matched points determine the 3D point by triangulation

   (this is an estimation problem)

The correspondence problem

Given a point x in one image find the corresponding point in the other image



This appears to be a 2D search problem, but it is reduced to a 1D search by the epipolar constraint

Challenges in Stereo Matching

（1）Distortion and noise

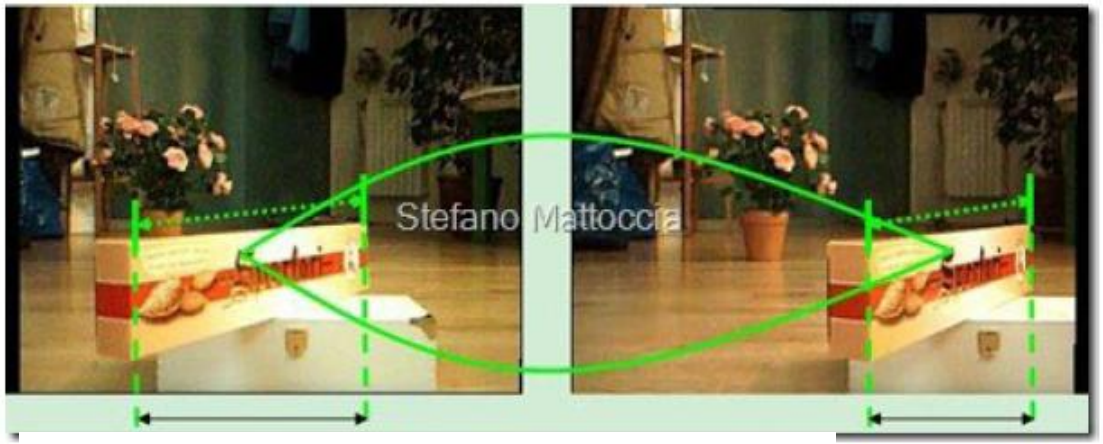


（2）Mirror reflection



Challenges in Stereo Matching

（3）Foreshortening



（4）Perspective distortions

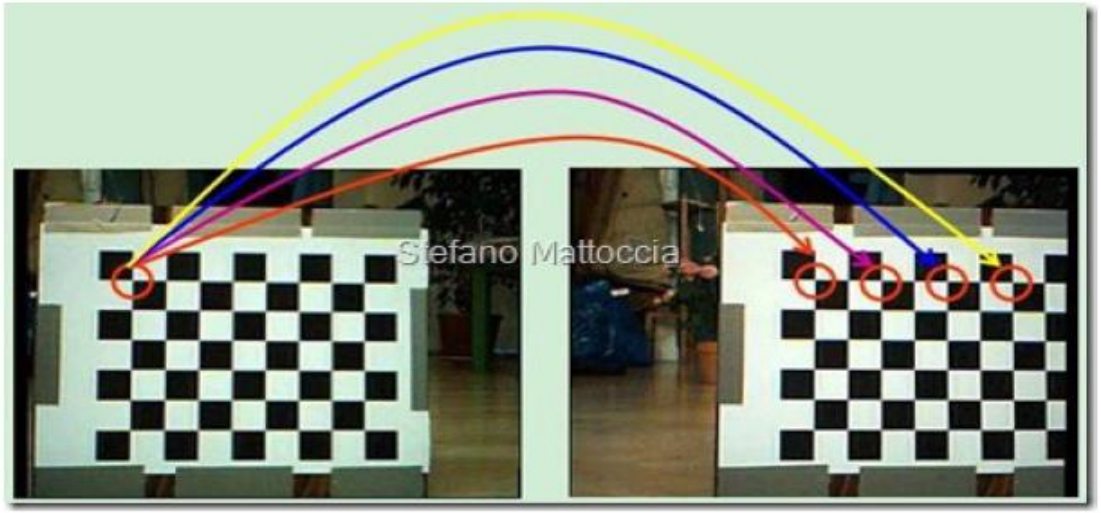


Challenges in Stereo Matching

（5）low texture



（6）repetitive/ambiguous patterns

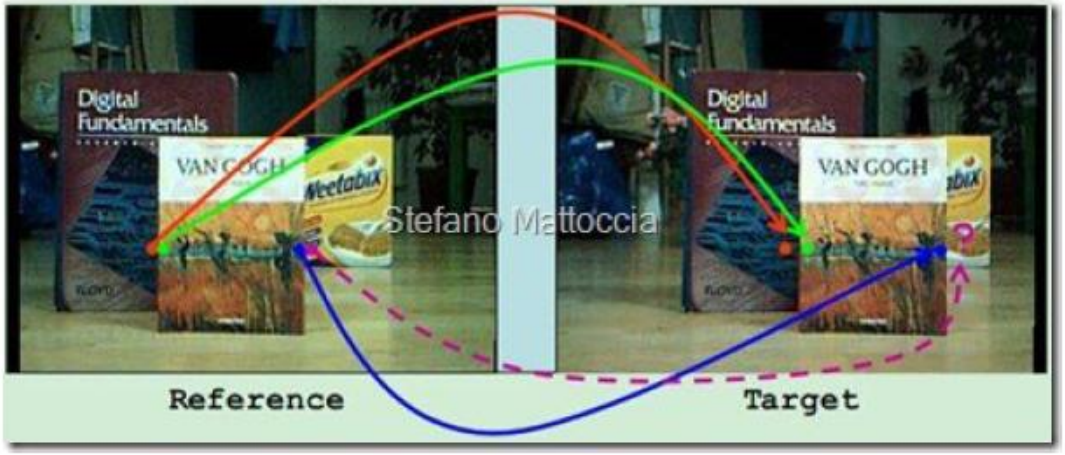


Challenges in Stereo Matching

（7）transparent object



（8）occlusion and discontinuous

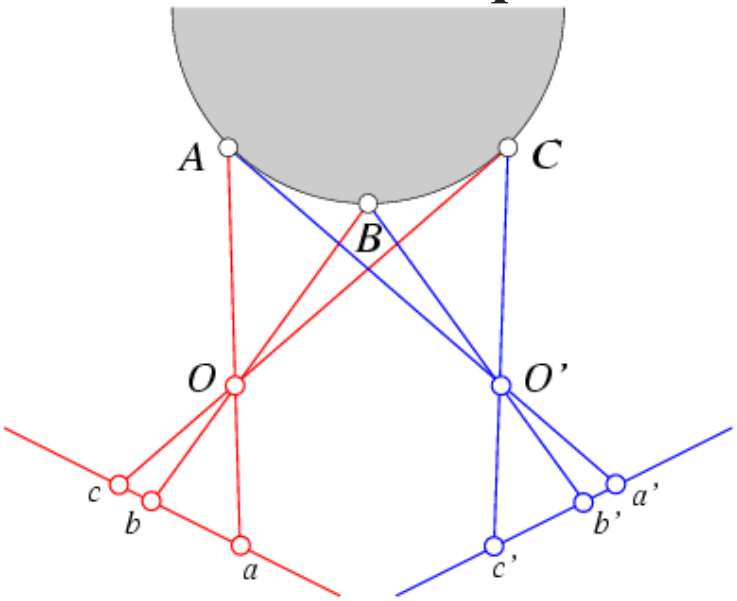


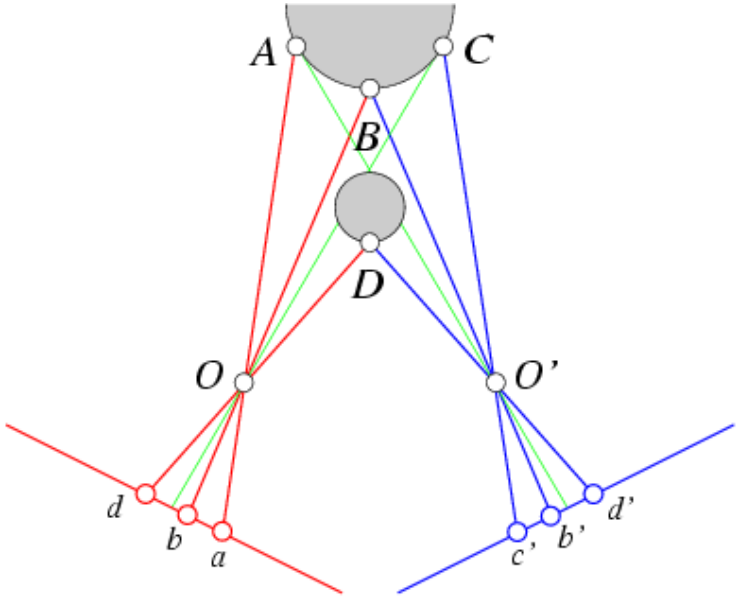
Constraint in Stereo Matching

1. Uniqueness Constraint
2. Continue Constraint
3. Feature Compatibility Constrain
4. Ordering Constraint
5. Mutual Correspondence Constraint
6. Disparity Limit Constraint
7. Epipolar Constraint

Constraint in Stereo Matching

1. Uniqueness Constraint

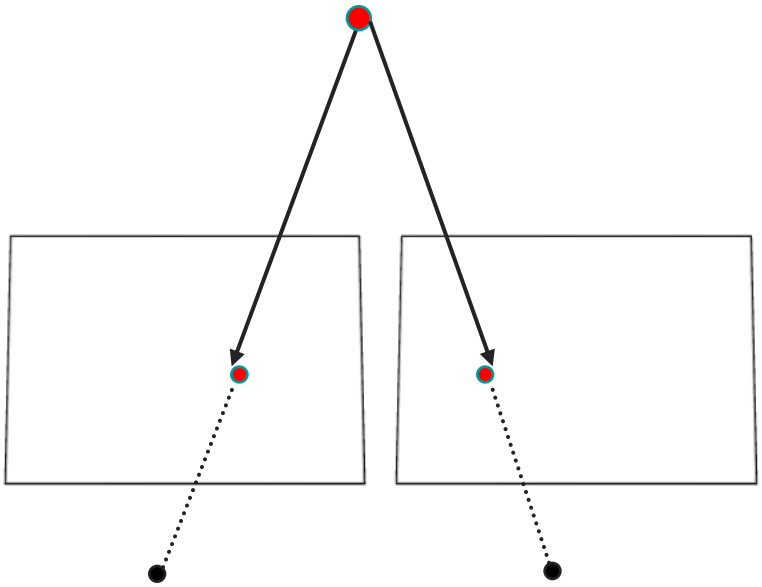




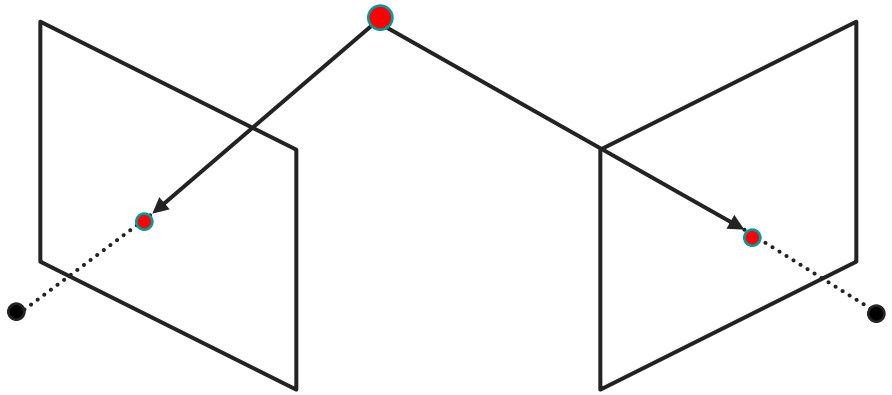
7. Epipolar Constraint

General case, with calibrated cameras

- The two cameras need not have parallel optical axes.

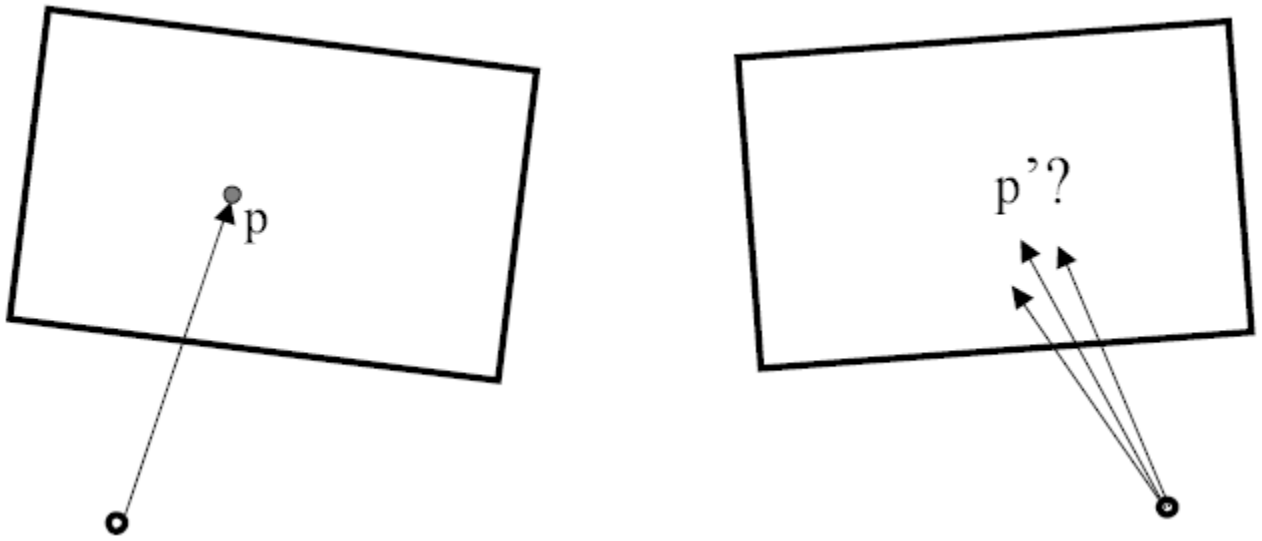


Vs.



Slide credit: Kristen Grauman

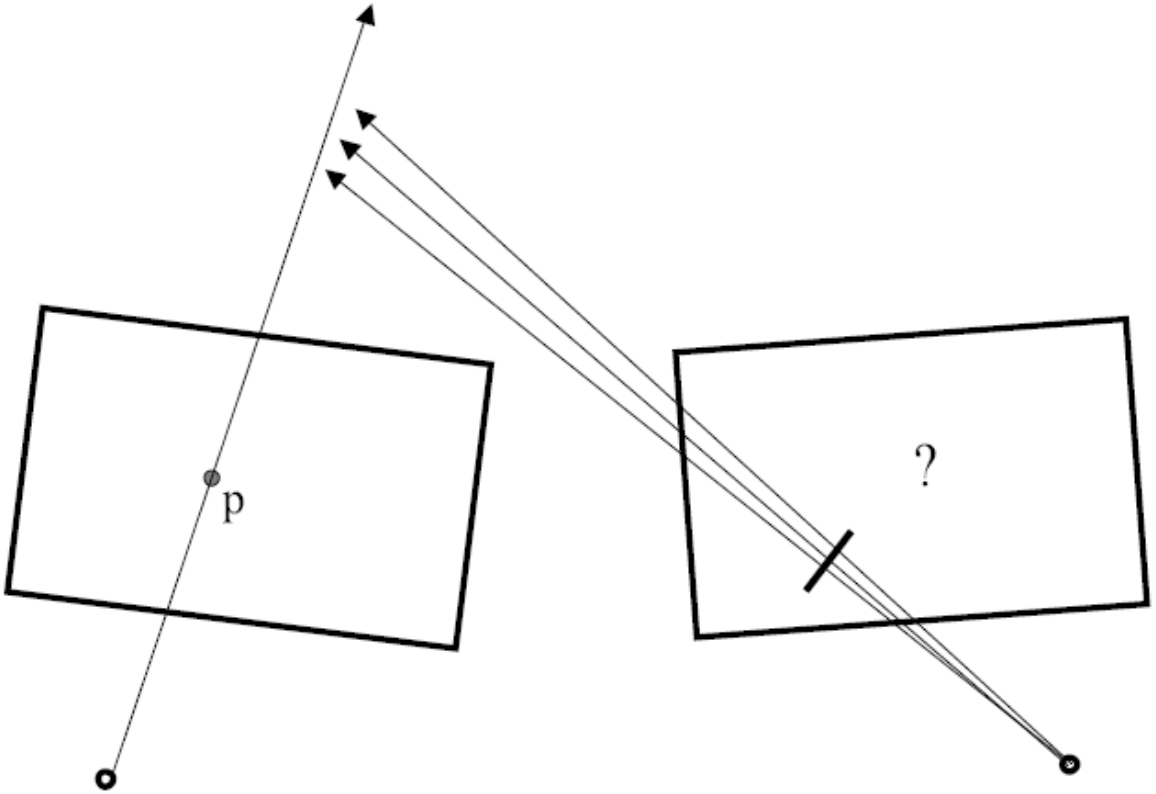
Stereo correspondence constraints



- **Given p in left image, where can corresponding point p' be?**

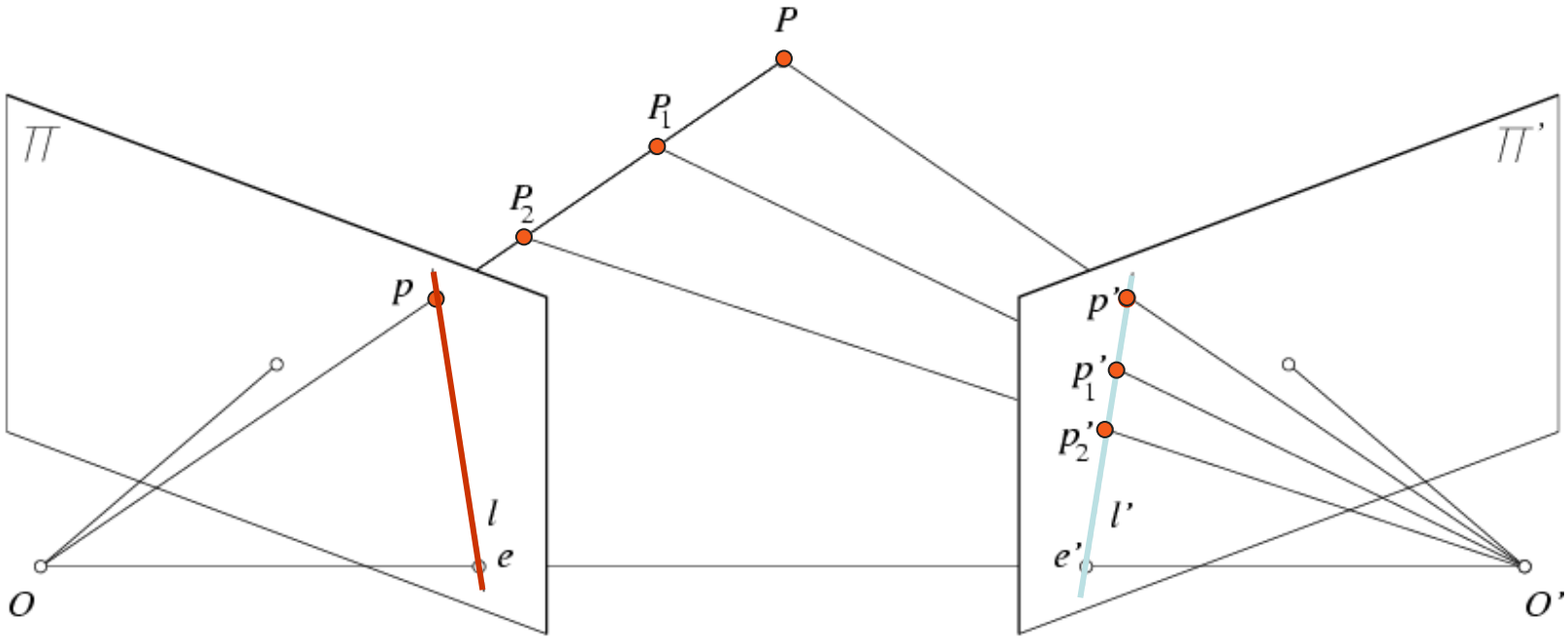
Slide credit: Kristen Grauman

Stereo correspondence constraints



Slide credit: Kristen Grauman

Epipolar constraint



Geometry of two views constrains where the corresponding pixel for some image point in the first view must occur in the second view:

•It must be on the line carved out by a plane connecting the world point and optical centers.

Why is this useful?

Slide credit: Kristen Grauman

Epipolar constraint

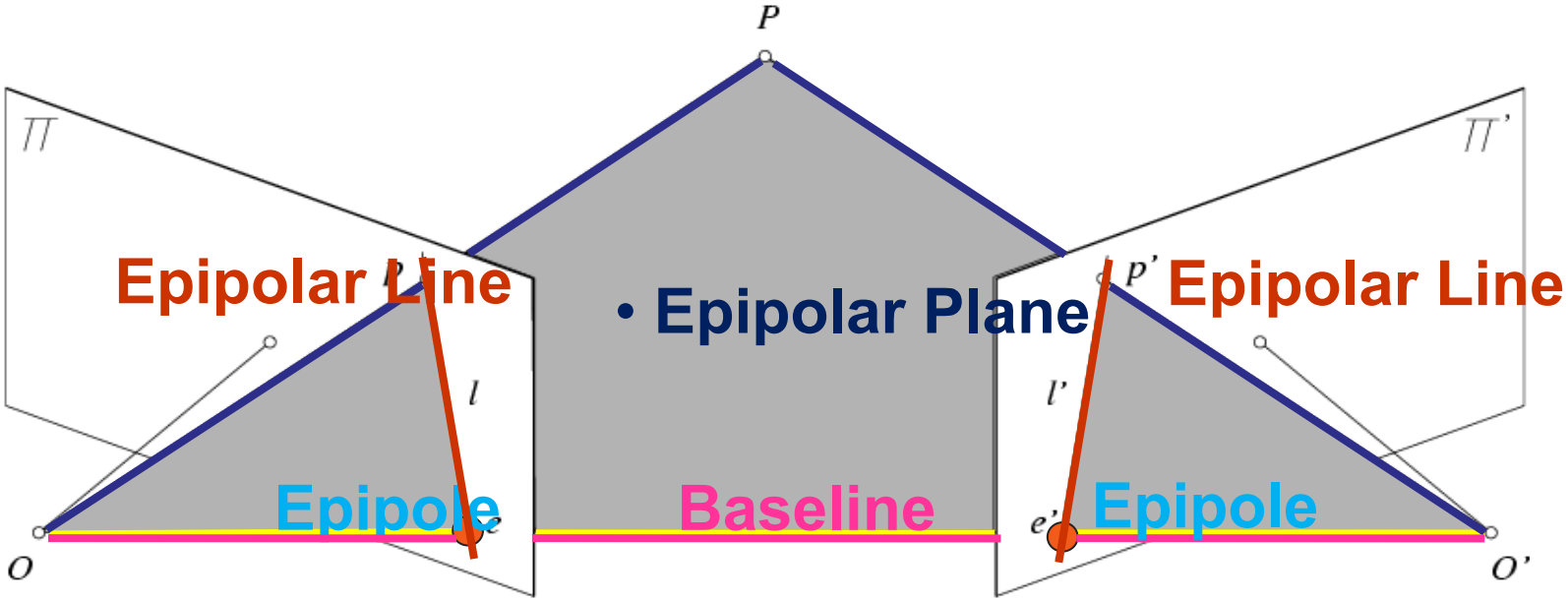


This is useful because it reduces the correspondence problem to a 1D search along an epipolar line.

Image from Andrew Zisserman

Slide credit: Kristen Grauman

Epipolar geometry

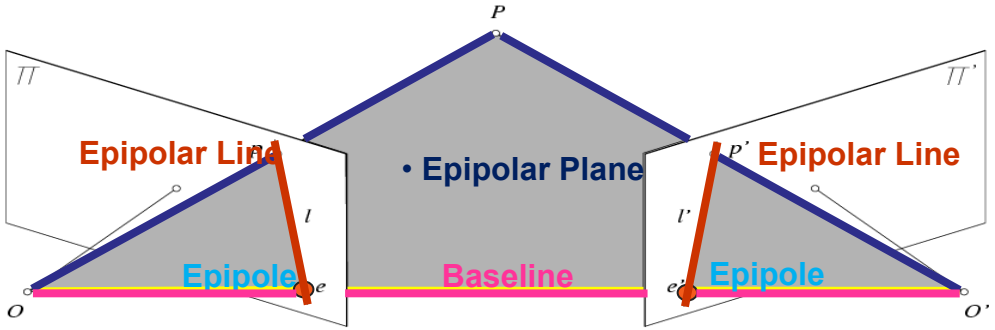


- **Baseline**: line joining the camera centers
- **Epipole**: point of intersection of baseline with the image plane
- **Epipolar plane**: plane containing baseline and world point
- **Epipolar line**: intersection of epipolar plane with the image plane
- All epipolar lines intersect at the epipole
- An epipolar plane intersects the left and right image planes in epipolar lines

http://www.ai.sri.com/~luong/research/Meta3DViewer/EpipolarGeo.html

Slide credit: Kristen Grauman

Epipolar constraint

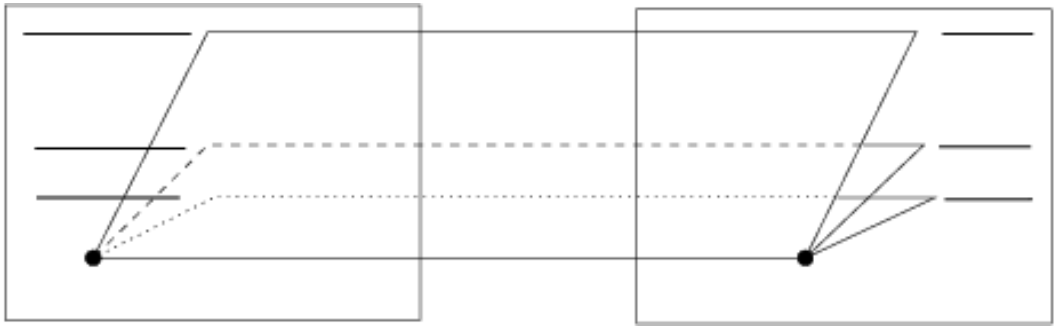


- Correspondence ploblem

  Simply the searching in 2D space to 1D



Example: parallel cameras



Where are the epipoles?

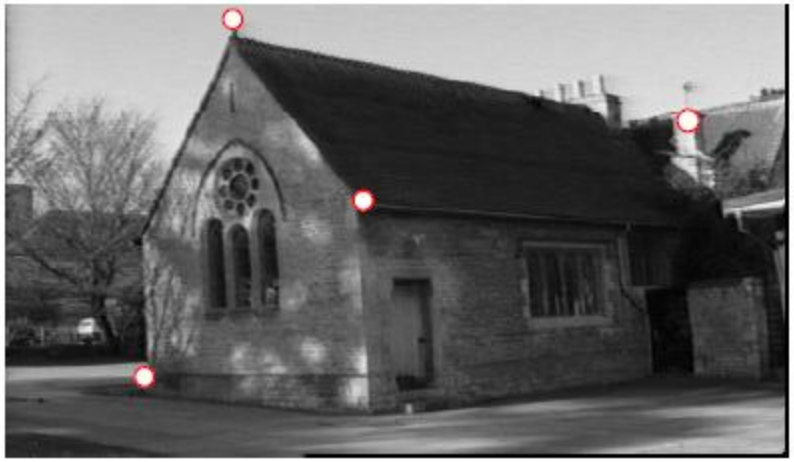
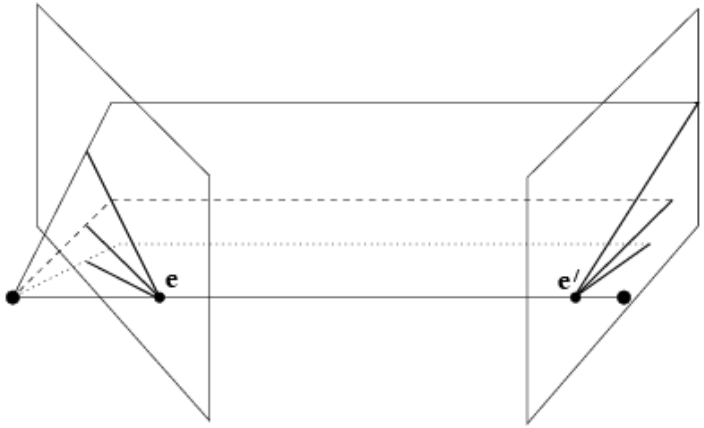




Figure from Hartley & Zisserman

Slide credit: Kristen Grauman

Example: converging cameras





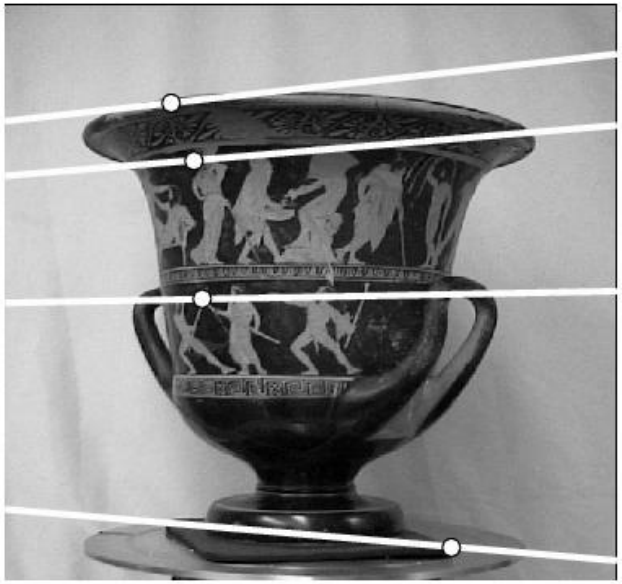
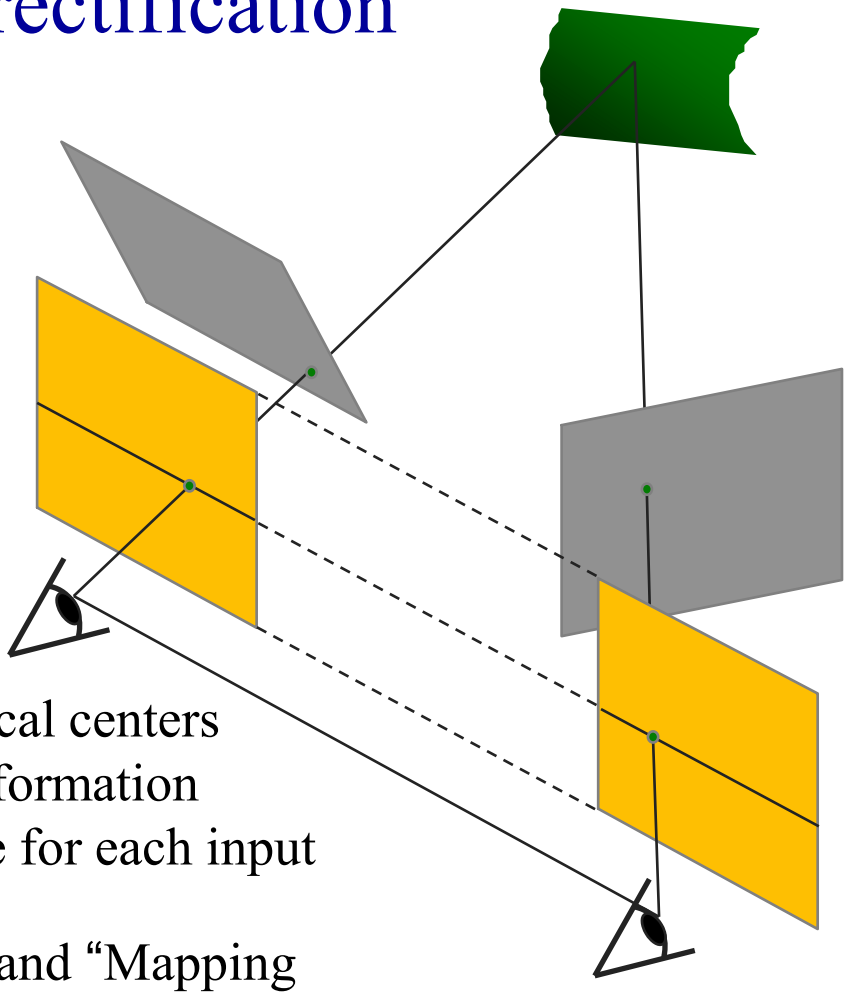


Figure from Hartley & Zisserman

Slide credit: Kristen Grauman

Stereo image rectification

In practice, it is convenient if image scanlines (rows) are the epipolar lines.



Reproject image planes onto a common plane parallel to the line between optical centers

Pixel motion is horizontal after this transformation

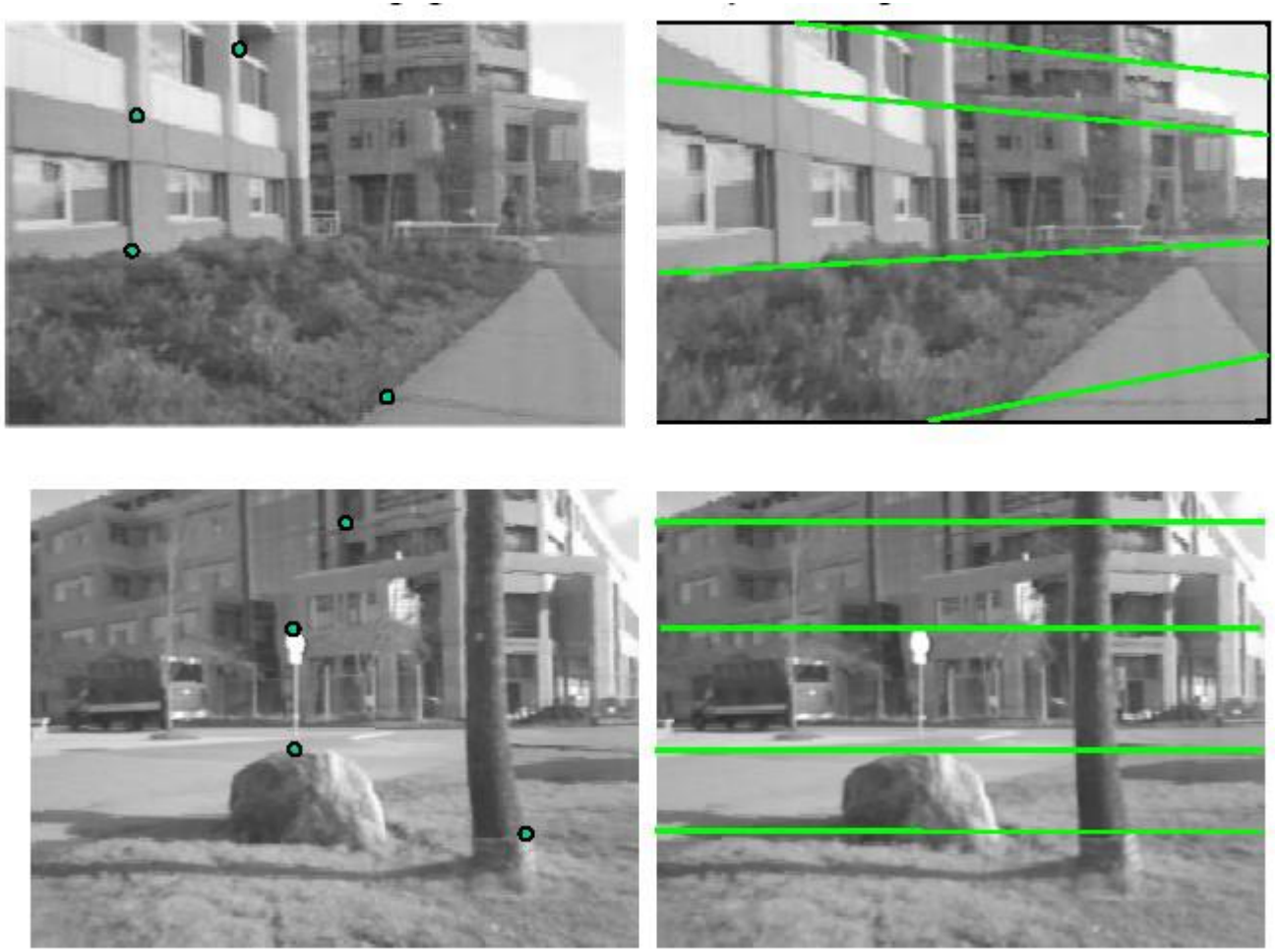
Two homographies (3x3 transforms), one for each input image reprojection

See Szeliski book, Sect. 2.1.5, Fig. 2.12, and "Mapping from one camera to another" p. 56

Adapted from Li Zhang

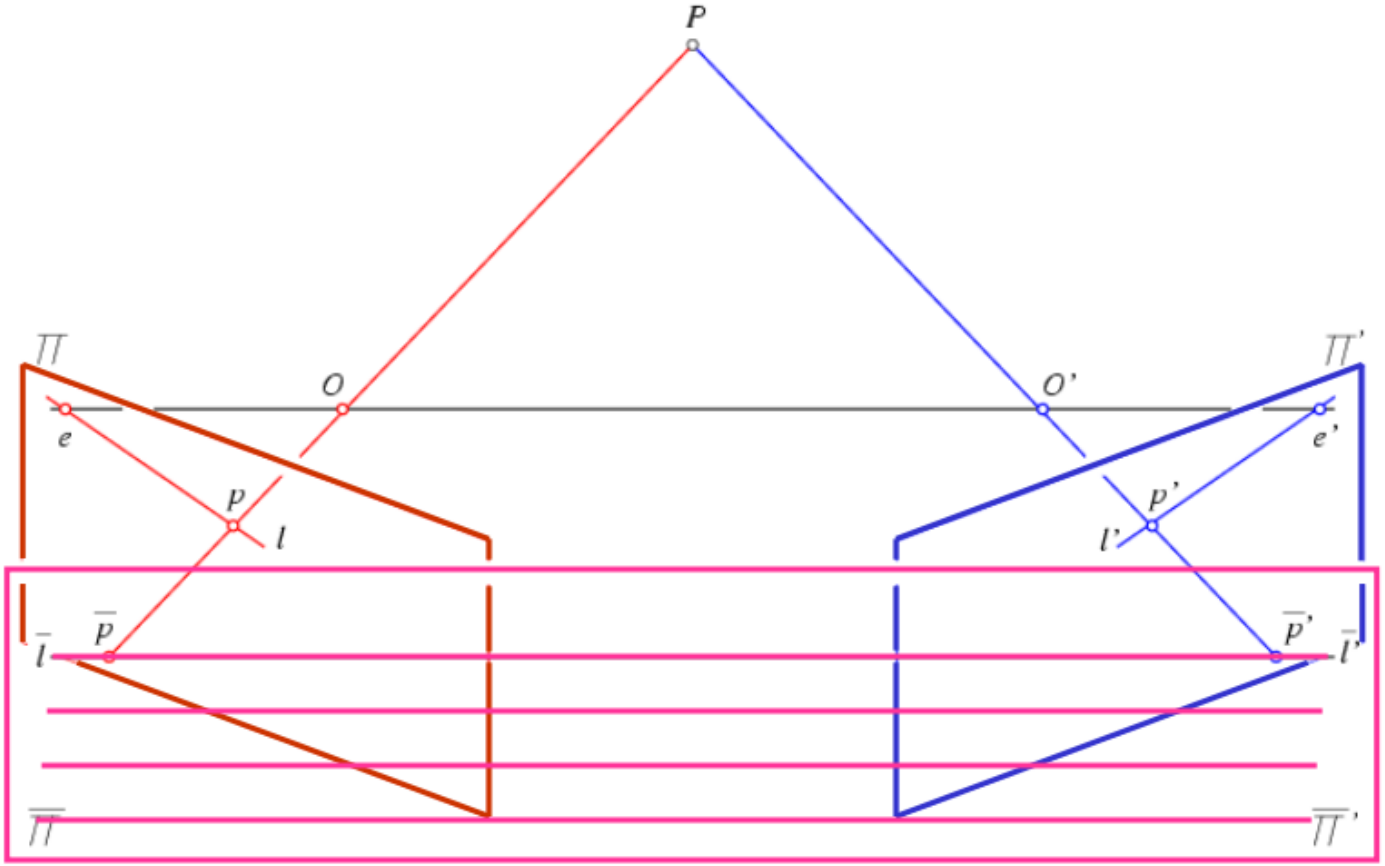
Slide credit: Kristen Grauman

Stereo image rectification



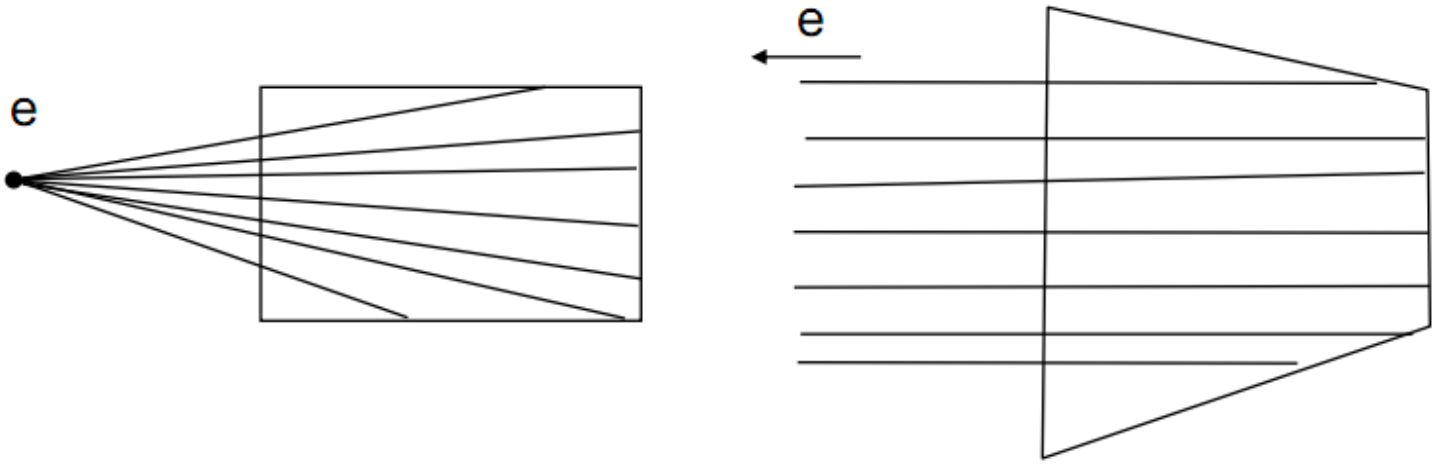
Slide credit: Kristen Grauman

Stereo image rectification



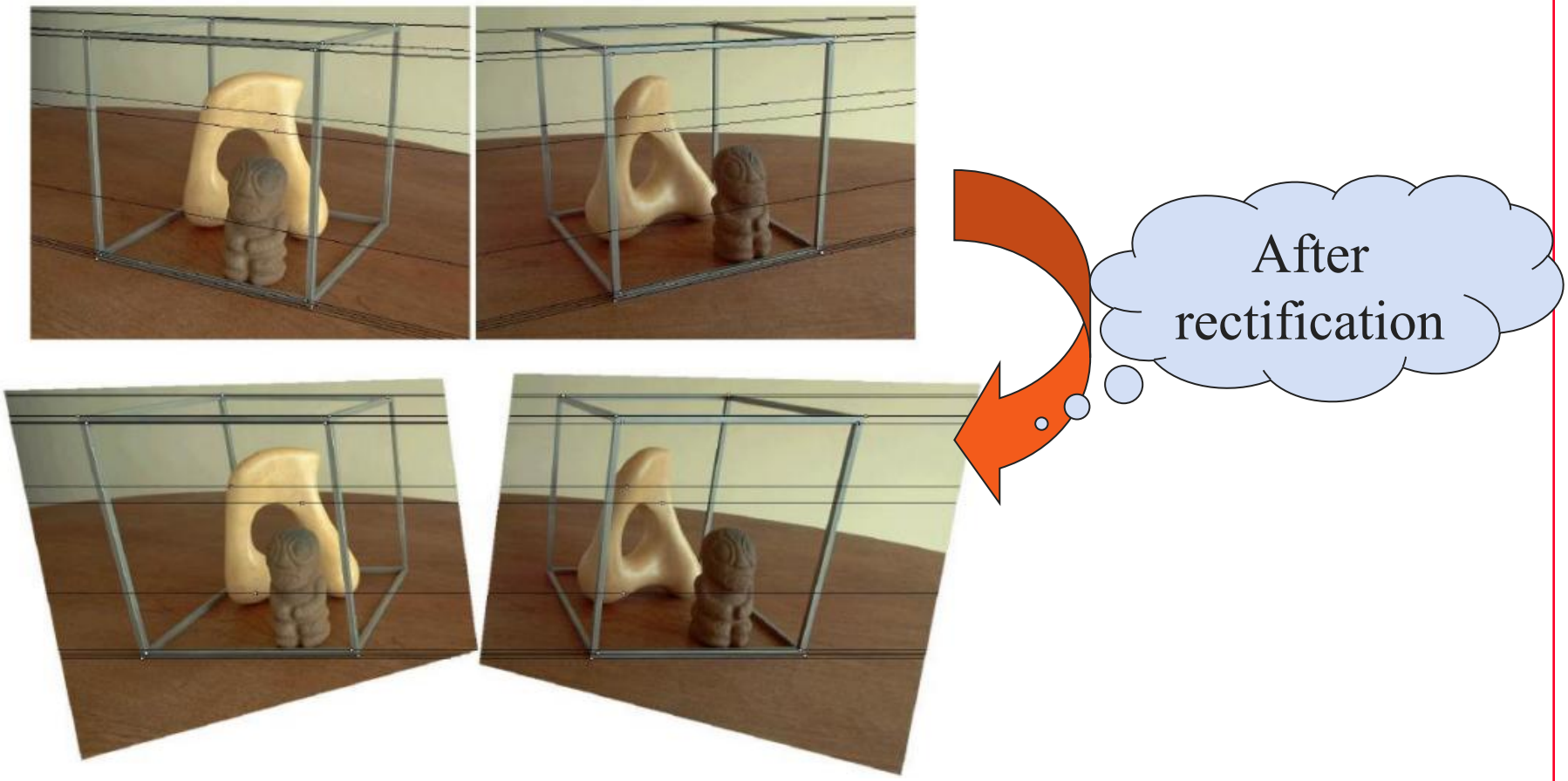
Stereo image rectification

Put the Epipole at infinity





Stereo image rectification



Agenda

- Depth perception
- Stereo vision
- Stereo Matching
- 3D Construction

Correspondence Problem

- Two classes of algorithms:

  Correlation-based algorithms

  Produce a DENSE set of correspondences

  Feature-based algorithms

  Produce a SPARSE set of correspondences

Stereo – Photometric Constraint

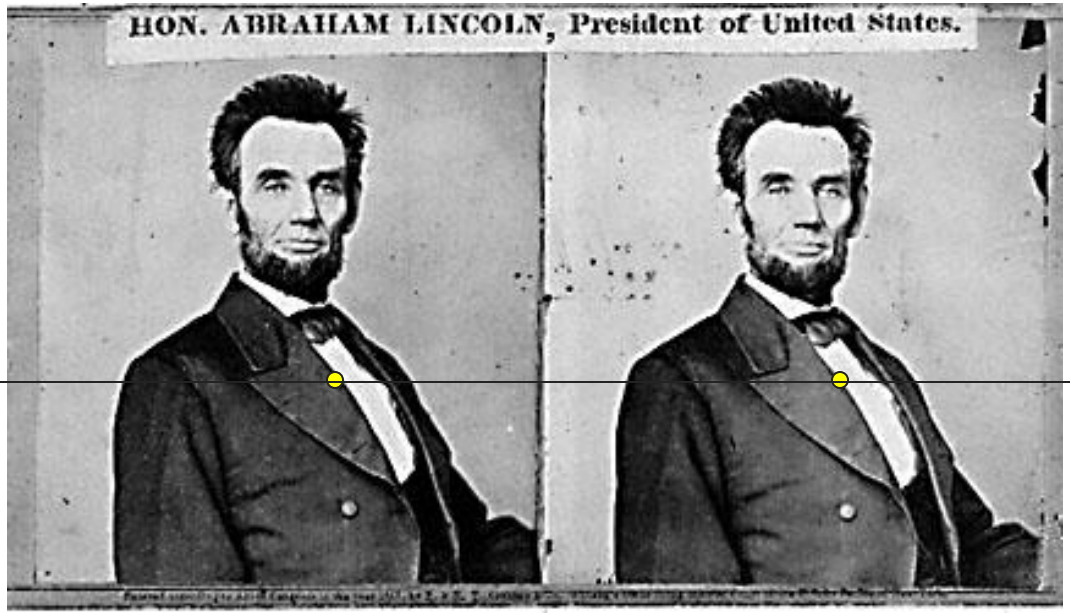
- Same world point has same intensity in both images.
  - Lambertian fronto-parallel
  - Issues (noise, specularities, foreshortening)





- Difficulties – ambiguities, large changes of appearance, due to change Of viewpoint, non-uniquess

Stereo Matching



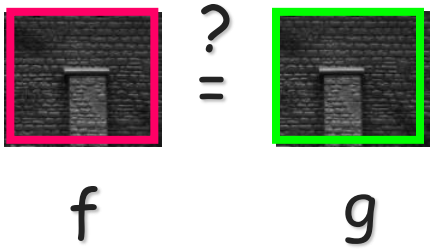
What if ?

For each scanline , for each pixel in the left image

- compare with every pixel on same epipolar line in right image
- pick pixel with minimum match cost
- This will never work, so: **improvement match windows**

(slides O. Camps)

Comparing Windows:



$$SSD = \sum_{[i,j]\in R} (f(i,j) - g(i,j))^2$$

$$C_{fg} = \sum_{[i,j]\in R} f(i,j)g(i,j)$$

Most popular

For each window, match to closest window on epipolar line in other image.

(slides O. Camps)

Comparing Windows:

Minimize $$\sum_{[i,j]\in R} (f(i,j) - g(i,j))^2$$ Sum of Squared Differences

Maximize $$C_{fg} = \sum_{[i,j]\in R} f(i,j)g(i,j)$$ Cross correlation

It is closely related to the SSD:

$$SSD = \sum_{[i,j]\in R} (f-g)^2 =$$

$$= \sum_{[i,j]\in R} f^2 + \sum_{[i,j]\in R} g^2 - 2\sum_{[i,j]\in R} fg$$

Region based Similarity Metrics

- Sum of squared differences

$$SSD(h) = \sum_{\tilde{\mathbf{x}} \in W(\mathbf{x})} \|I_1(\tilde{\mathbf{x}}) - I_2(h(\tilde{\mathbf{x}})\|^2$$

- Normalize cross-correlation

$$NCC(h) = \frac{\sum_{W(\mathbf{x})} (I_1(\tilde{\mathbf{x}}) - \bar{I}_1)(I_2(h(\tilde{\mathbf{x}})) - \bar{I}_2))}{\sqrt{\sum_{W(\mathbf{x})} (I_1(\tilde{\mathbf{x}}) - \bar{I}_1)^2 \sum_{W(\mathbf{x})} (I_2(h(\tilde{\mathbf{x}})) - \bar{I}_2)^2)}}$$

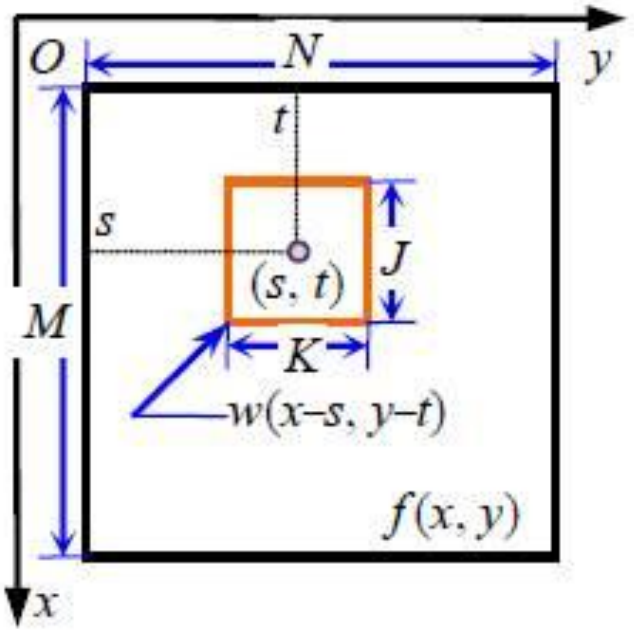
- Sum of absolute differences

$$SAD(h) = \sum_{\tilde{\mathbf{x}} \in W(\mathbf{x})} |I_1(\tilde{\mathbf{x}}) - I_2(h(\tilde{\mathbf{x}}))|$$

NCC function

CC function

$$c(s,t)=\sum_x\sum_y f(x,y)w(x-s,y-t)$$



correlation coefficient

$$\gamma(s,t)=\frac{\sum_x\sum_y\left[f(x,y)-\bar{f}(x,y)\right]\left[w(x-s,y-t)-\bar{w}\right]}{\left\{\sum_x\sum_y\left[f(x,y)-\bar{f}(x,y)\right]^2\sum_x\sum_y\left[w(x-s,y-t)-\bar{w}\right]^2\right\}^{1/2}}$$

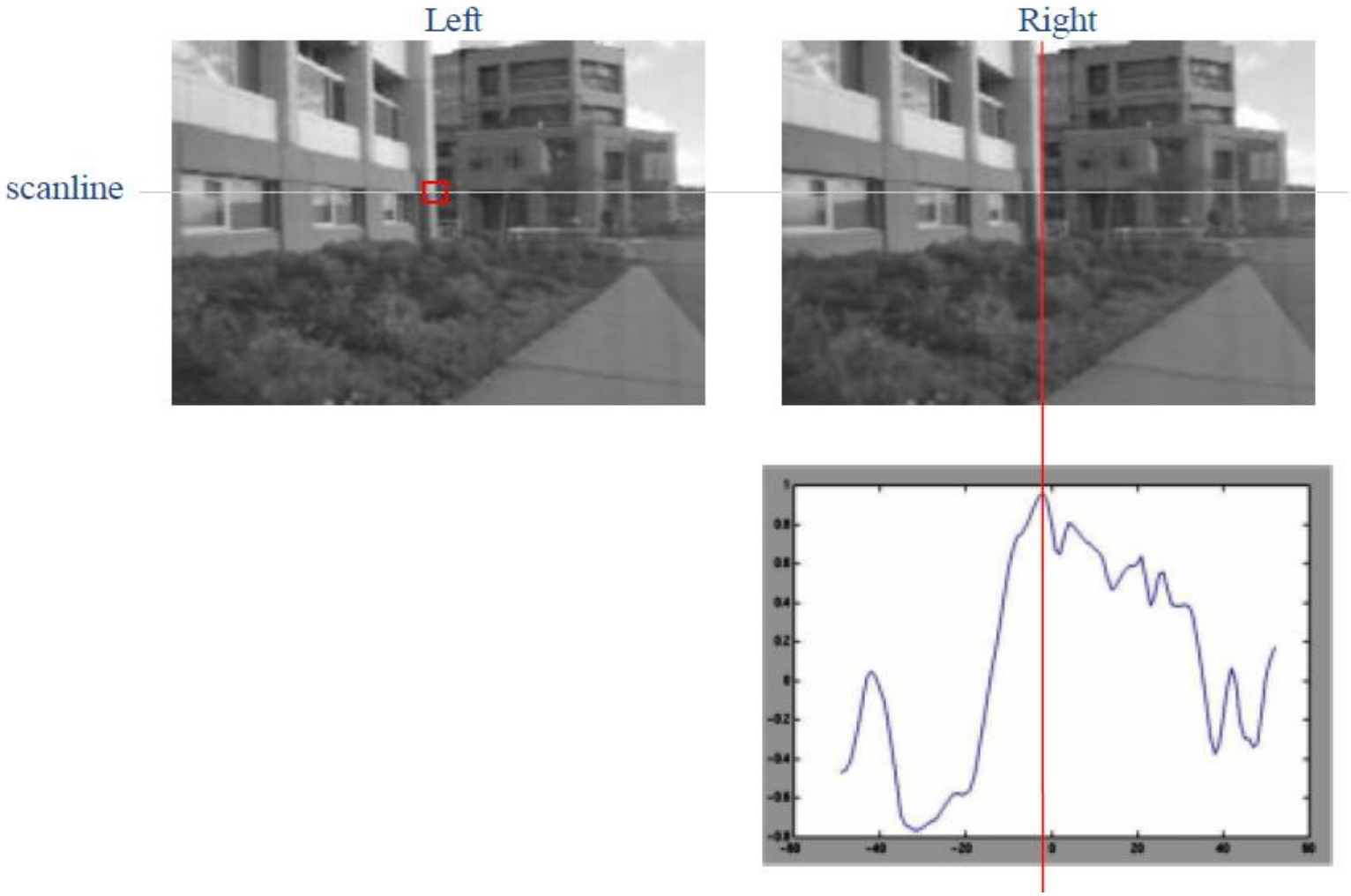
NCC score for two widely separated views



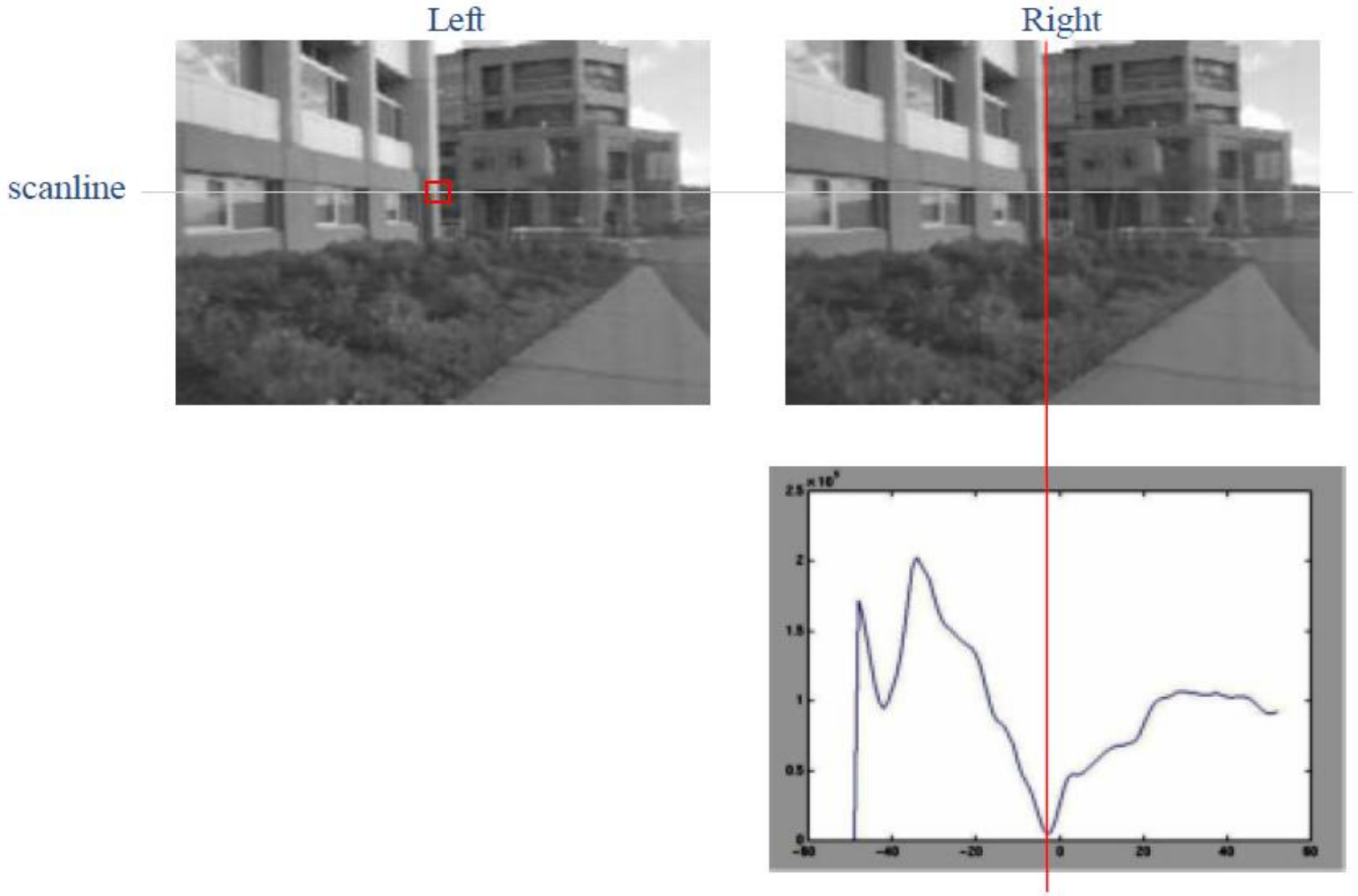


NCC score

NCC



Sum of squared differences

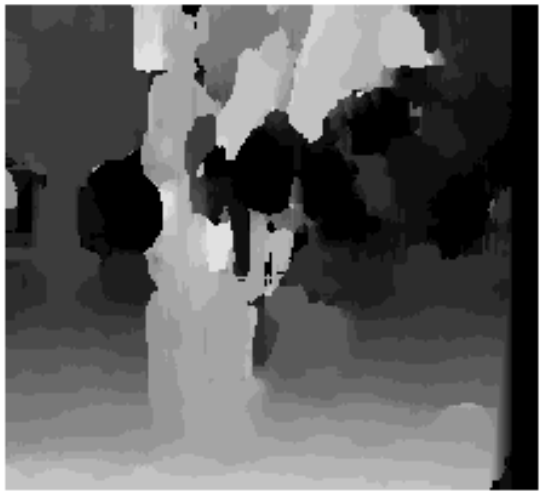


Window size





W = 3



W = 20

- Effect of window size

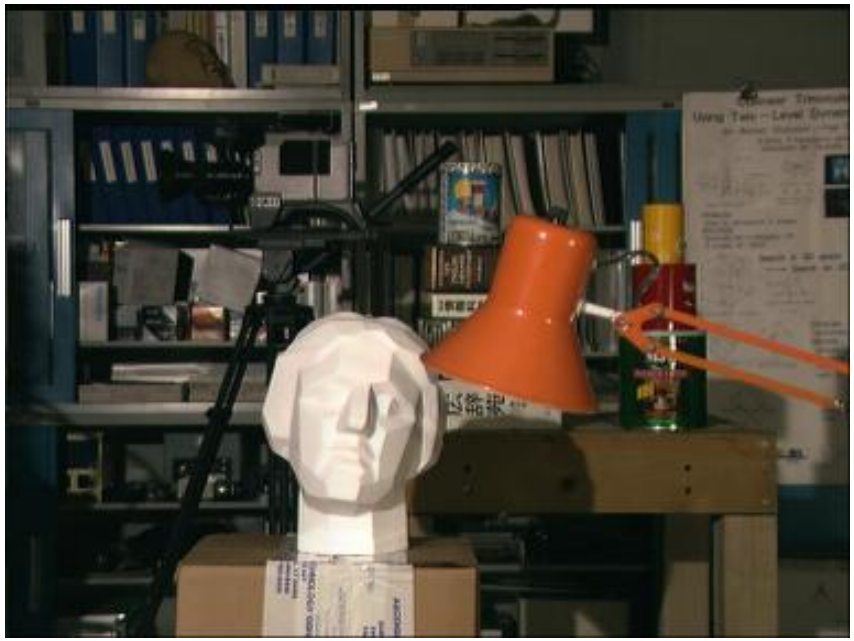


Better results with *adaptive window*

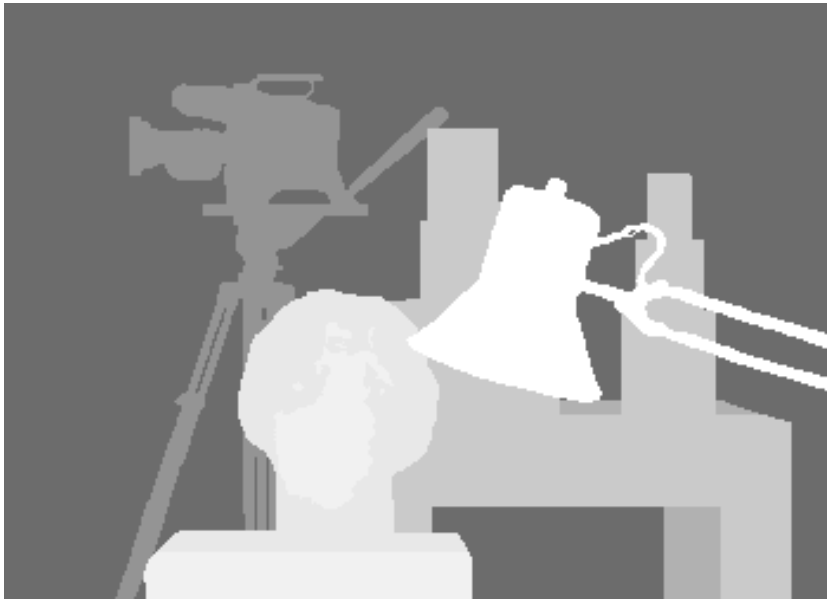
- T. Kanade and M. Okutomi, *A Stereo Matching Algorithm with an Adaptive Window: Theory and Experiment*,, Proc. International Conference on Robotics and Automation, 1991.
- D. Scharstein and R. Szeliski. Stereo matching with nonlinear diffusion. International Journal of Computer Vision, 28(2):155-174, July 1998

Stereo results

Data from University of Tsukuba



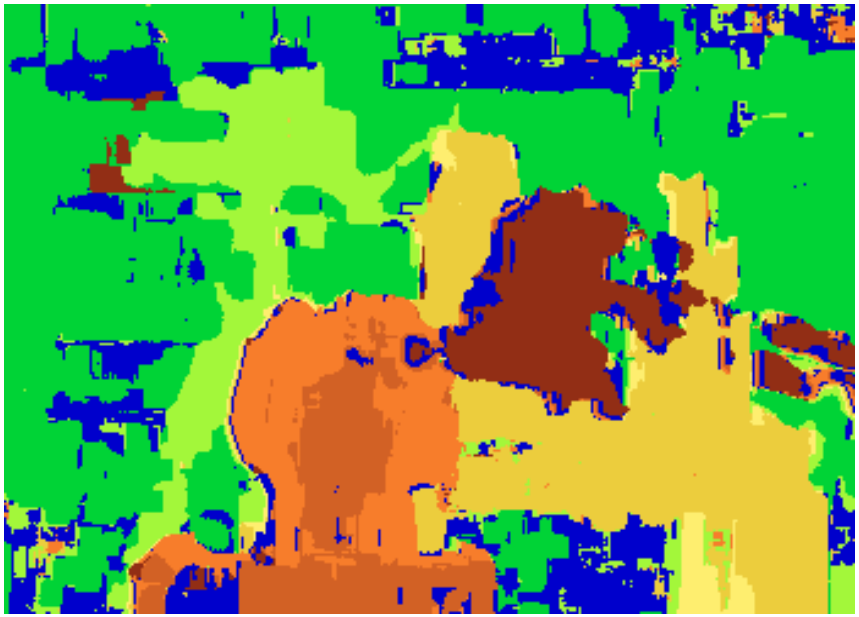
Scene



Ground truth

(Seitz)

Results with window correlation



Window-based matching
(best window size)



Ground truth

(Seitz)

Results with better method



State of the art



Ground truth

Boykov et al., Fast Approximate Energy Minimization via Graph Cuts, International Conference on Computer Vision, September 1999.

(Seitz)

Example image pair – parallel cameras





First image



Second image



Dense correspondence algorithm

Parallel camera example – epipolar lines are corresponding rasters





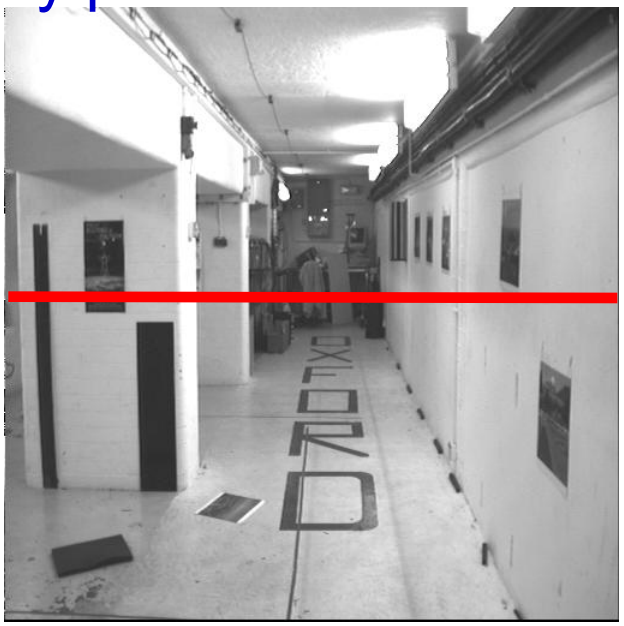
epipolar line

Search problem (geometric constraint): for each point in the left image, the corresponding point in the right image lies on the epipolar line (1D ambiguity)

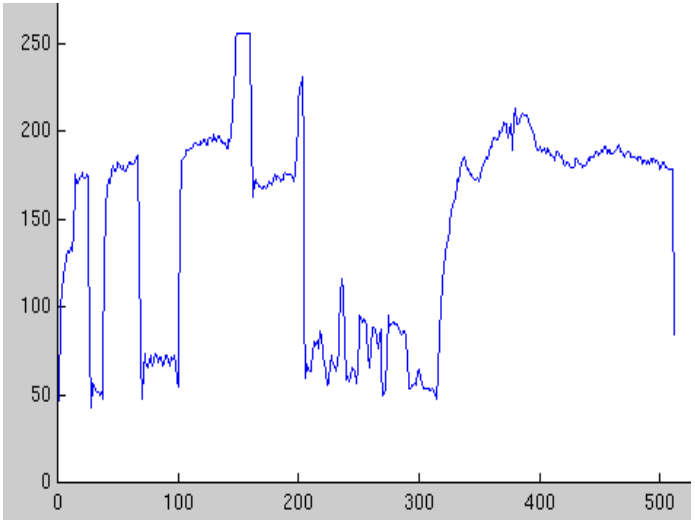
Disambiguating assumption (photometric constraint): the intensity neighbourhood of corresponding points are similar across images

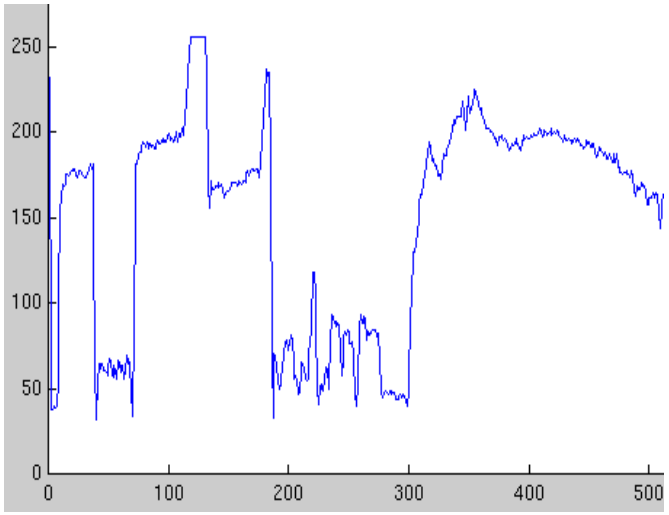
Measure similarity of neighbourhood intensity by cross-correlation

Intensity profiles









- Clear correspondence between intensities, but also noise and ambiguity

Normalized Cross Correlation

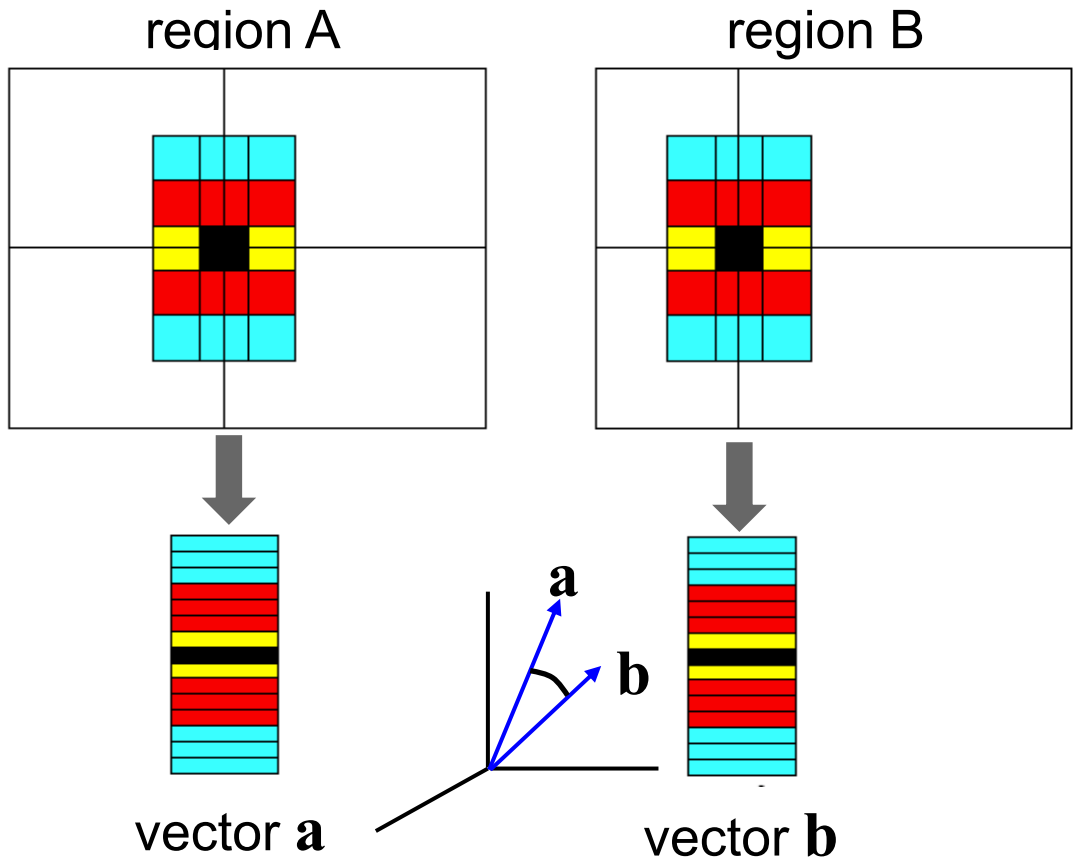
$$\mathrm{NCC} = \frac{\sum_i \sum_j A(i,j)B(i,j)}{\sqrt{\sum_i \sum_j A(i,j)^2}\sqrt{\sum_i \sum_j B(i,j)^2}}$$

write regions as vectors

$$\mathrm{A} \rightarrow \mathbf{a}, \quad \mathrm{B} \rightarrow \mathbf{b}$$

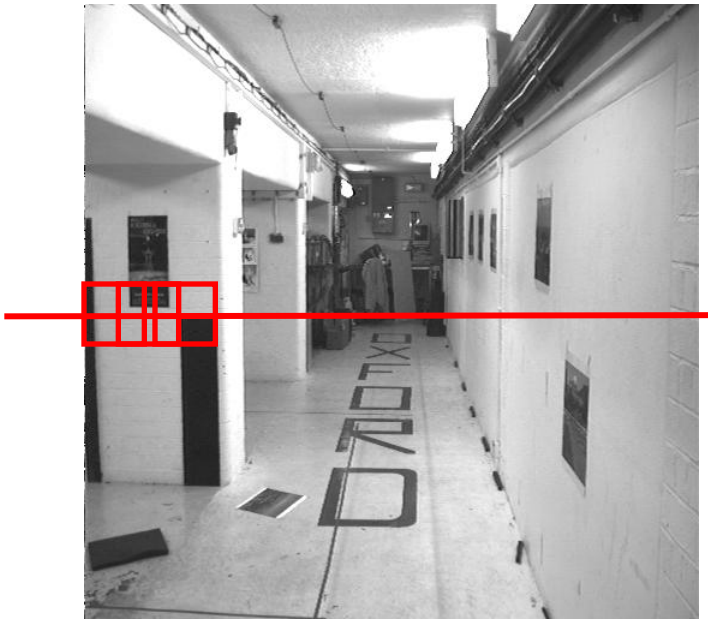
$$\mathrm{NCC} = \frac{\mathbf{a}.\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

$$-1 \leq \mathrm{NCC} \leq 1$$



Cross-correlation of neighbourhood regions





regions $\mathtt{A}, \mathtt{B}$, write as vectors $\mathbf{a}, \mathbf{b}$

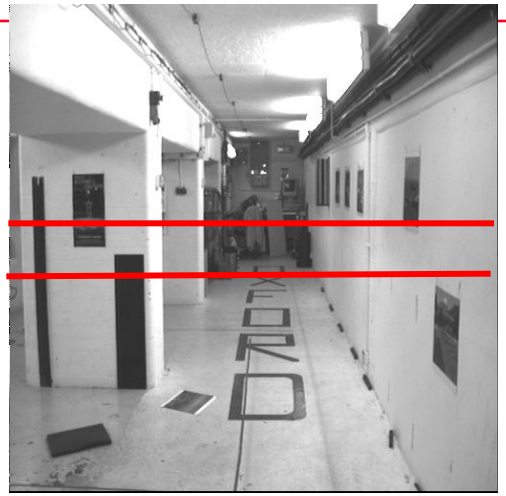
translate so that mean is zero

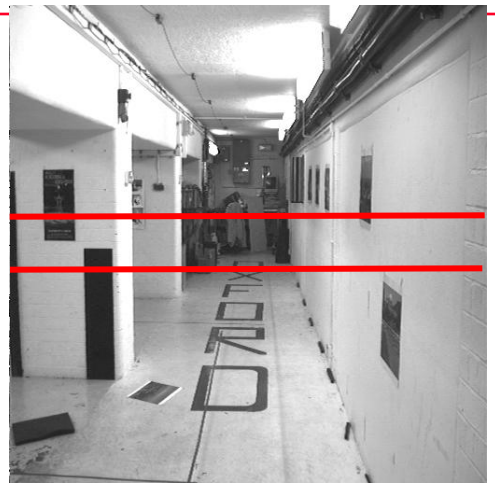
$$\mathtt{a} \rightarrow \mathtt{a} - \langle \mathbf{a} \rangle, \quad \mathtt{b} \rightarrow \mathtt{b} - \langle \mathbf{b} \rangle$$

$$\text{cross correlation} = \frac{\mathbf{a}.\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

Invariant to $I \rightarrow \alpha I + \beta$

(exercise)

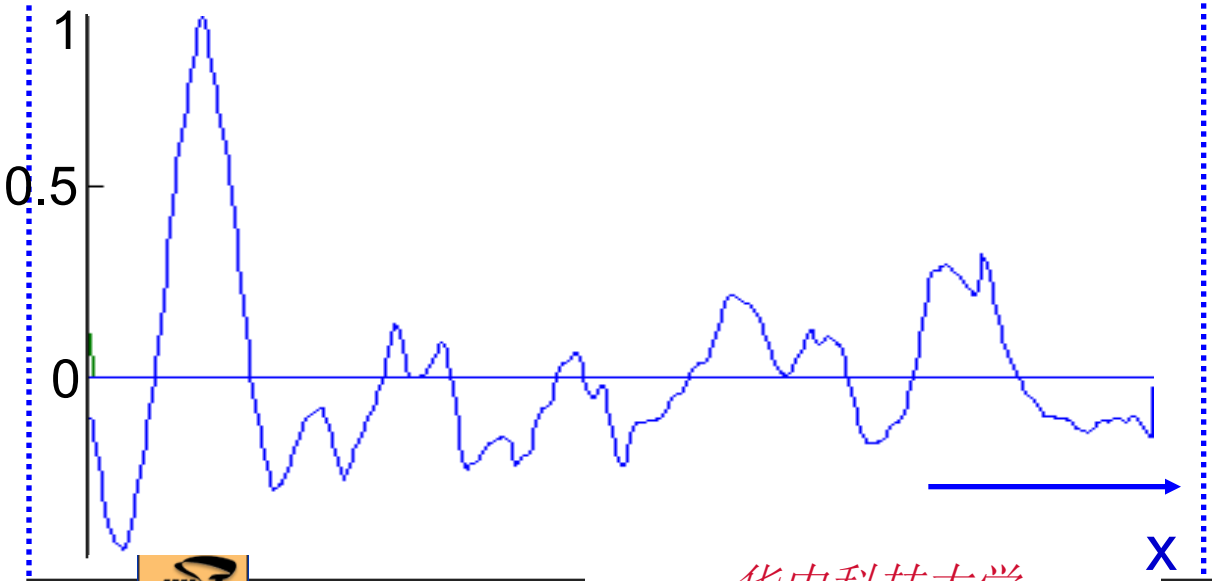




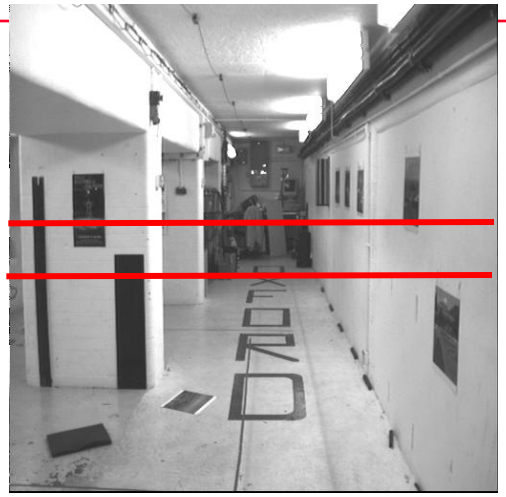


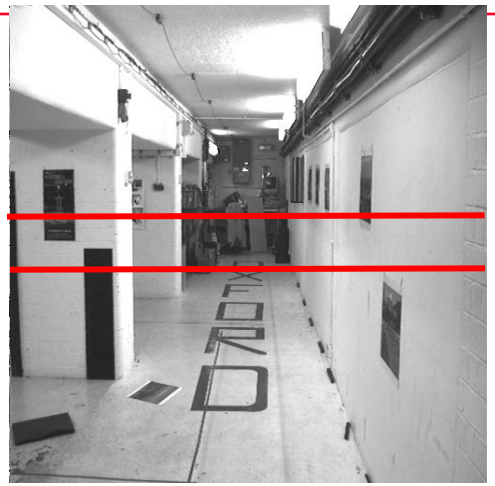
left image band

right image band



cross correlation



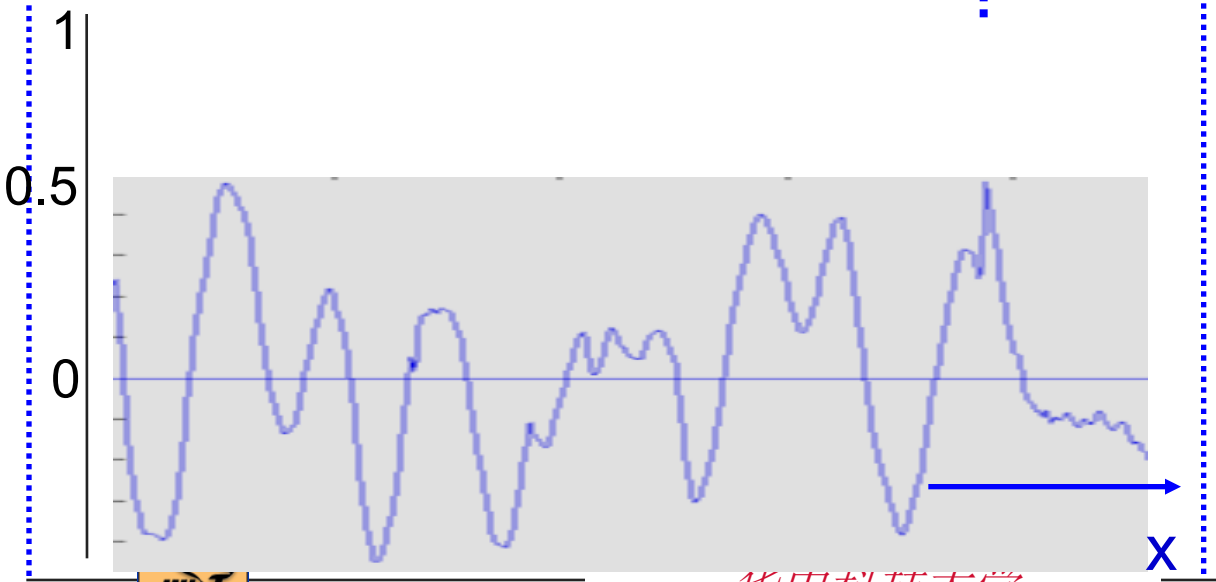


target region



left image band

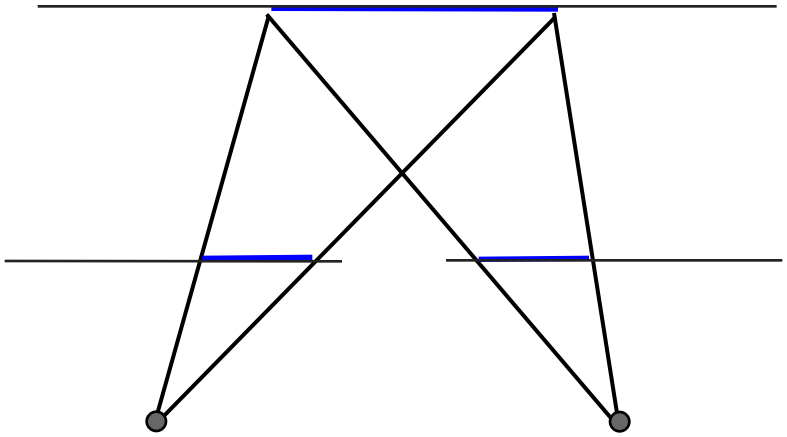
right image band



cross correlation

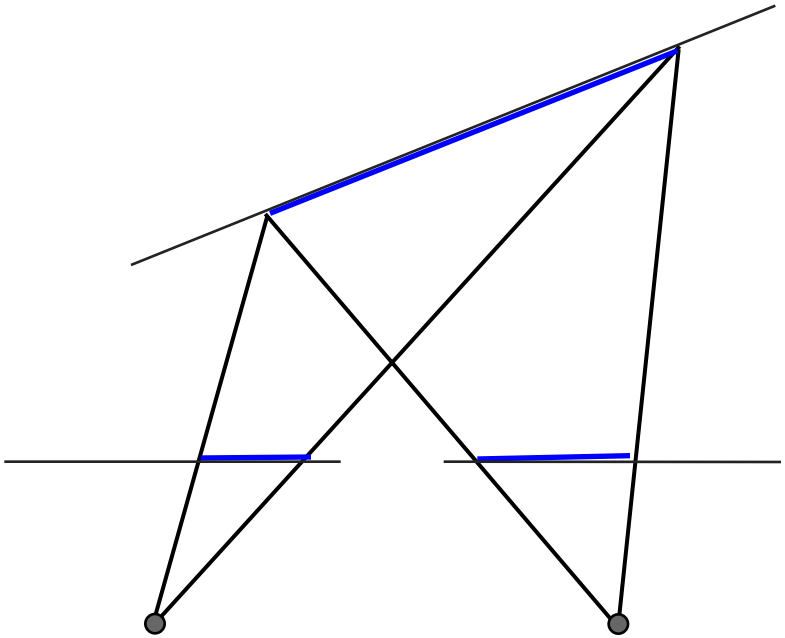
Why is cross-correlation such a poor measure in the second case?

1. The neighbourhood region does not have a “distinctive” spatial intensity distribution
2. Foreshortening effects



fronto-parallel surface

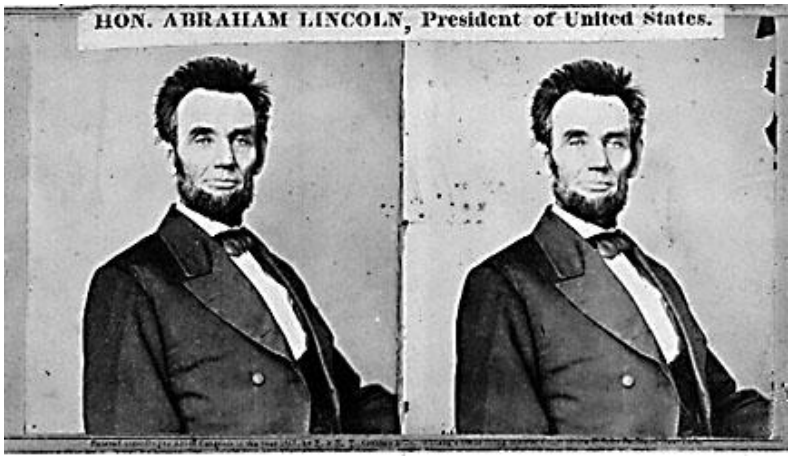
imaged length the same



slanting surface

imaged lengths differ

Limitations of similarity constraint



Textureless surfaces



Occlusions, repetition



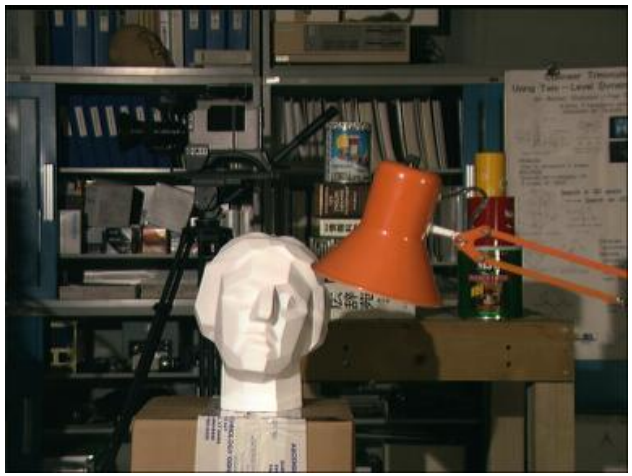




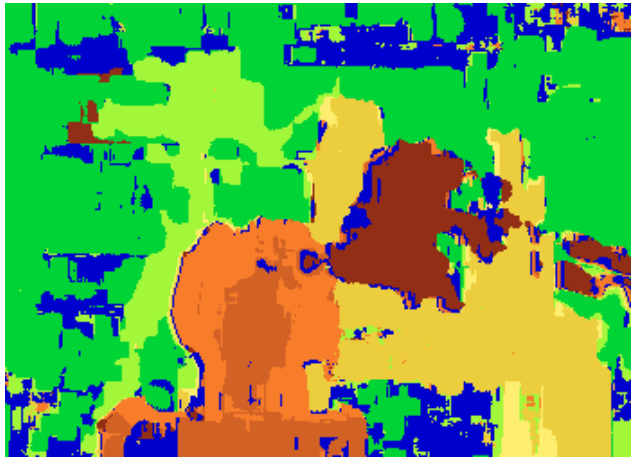
Non-Lambertian surfaces, specularities

Results with window search

Data



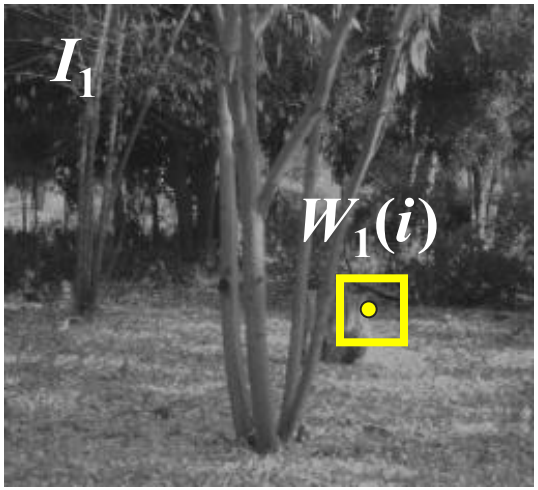
Window-based matching

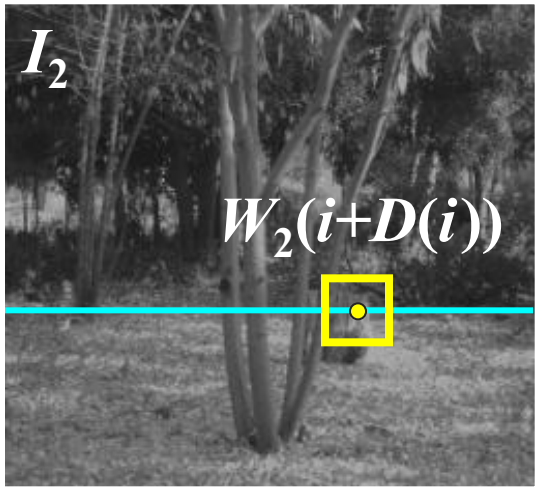


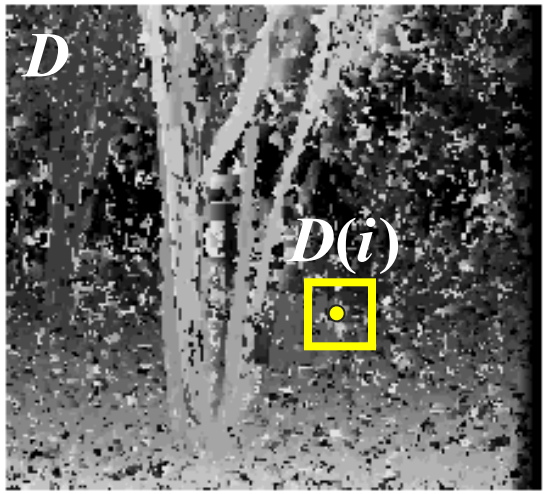
Ground truth



Stereo matching as energy minimization







MAP estimate of disparity image D: $P(D \mid I_1, I_2) \propto P(I_1, I_2 \mid D) P(D)$

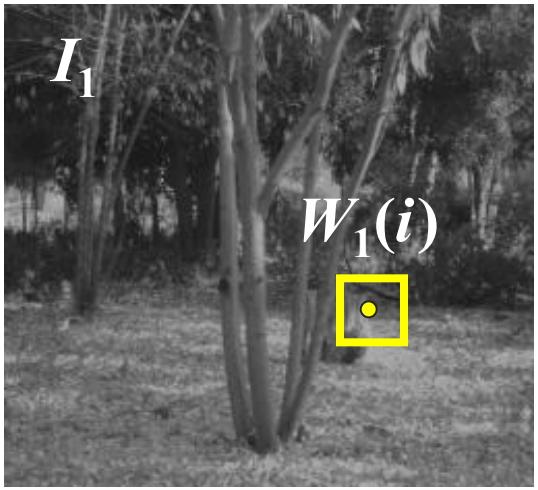
$$-\log P(D \mid I_1, I_2) \propto -\log P(I_1, I_2 \mid D) - \log P(D)$$

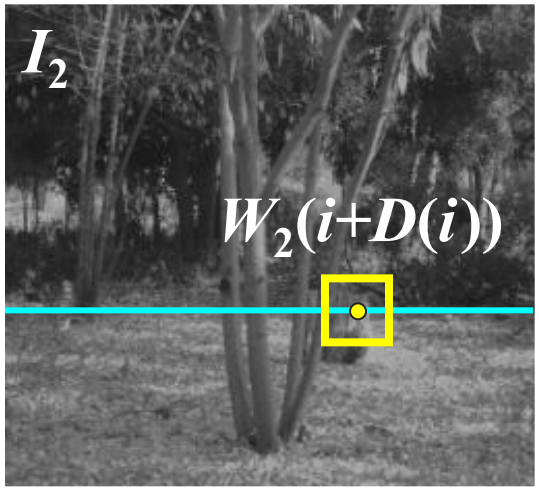
$$E = \alpha E_{\text{data}}(I_1, I_2, D) + \beta E_{\text{smooth}}(D)$$

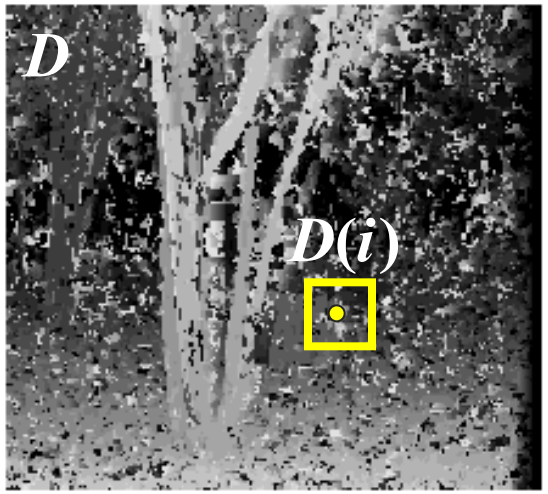
$$E_{\text{data}} = \sum_i \left(W_1(i) - W_2(i + D(i))\right)^2$$

$$E_{\text{smooth}} = \sum_{\text{neighbors } i,j} \rho\left(D(i) - D(j)\right)$$

Stereo matching as energy minimization







$$E = \alpha E_{\text{data}}(I_1, I_2, D) + \beta E_{\text{smooth}}(D)$$

$$E_{\text{data}} = \sum_i \left(W_1(i) - W_2(i + D(i))\right)^2$$

$$E_{\text{smooth}} = \sum_{\text{neighbors } i,j} \rho\left(D(i) - D(j)\right)$$

- Energy functions of this form can be minimized using *graph cuts*

Y. Boykov, O. Veksler, and R. Zabih, Fast Approximate Energy Minimization via Graph Cuts, PAMI 2001

Graph cuts solution



Graph cuts

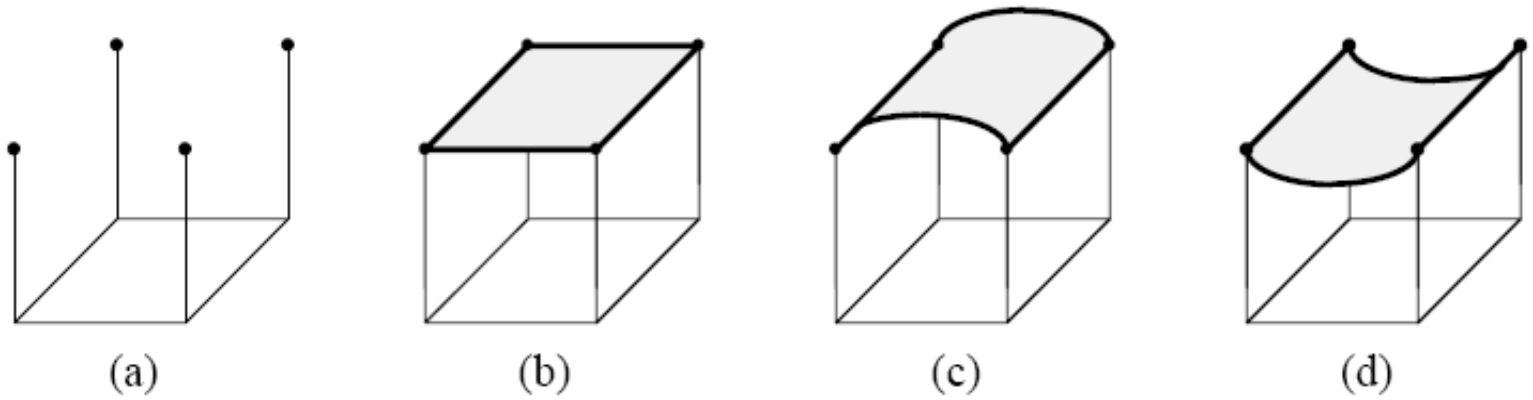


Ground truth

Y. Boykov, O. Veksler, and R. Zabih, Fast Approximate Energy Minimization via Graph Cuts, PAMI 2001

For the latest and greatest: http://www.middlebury.edu/stereo/

- Feature based stereo matching



vision.middlebury.edu

stereo • mview • MRF • flow

Stereo

Evaluation • Datasets • Code • Submit

Daniel Scharstein • Richard Szeliski



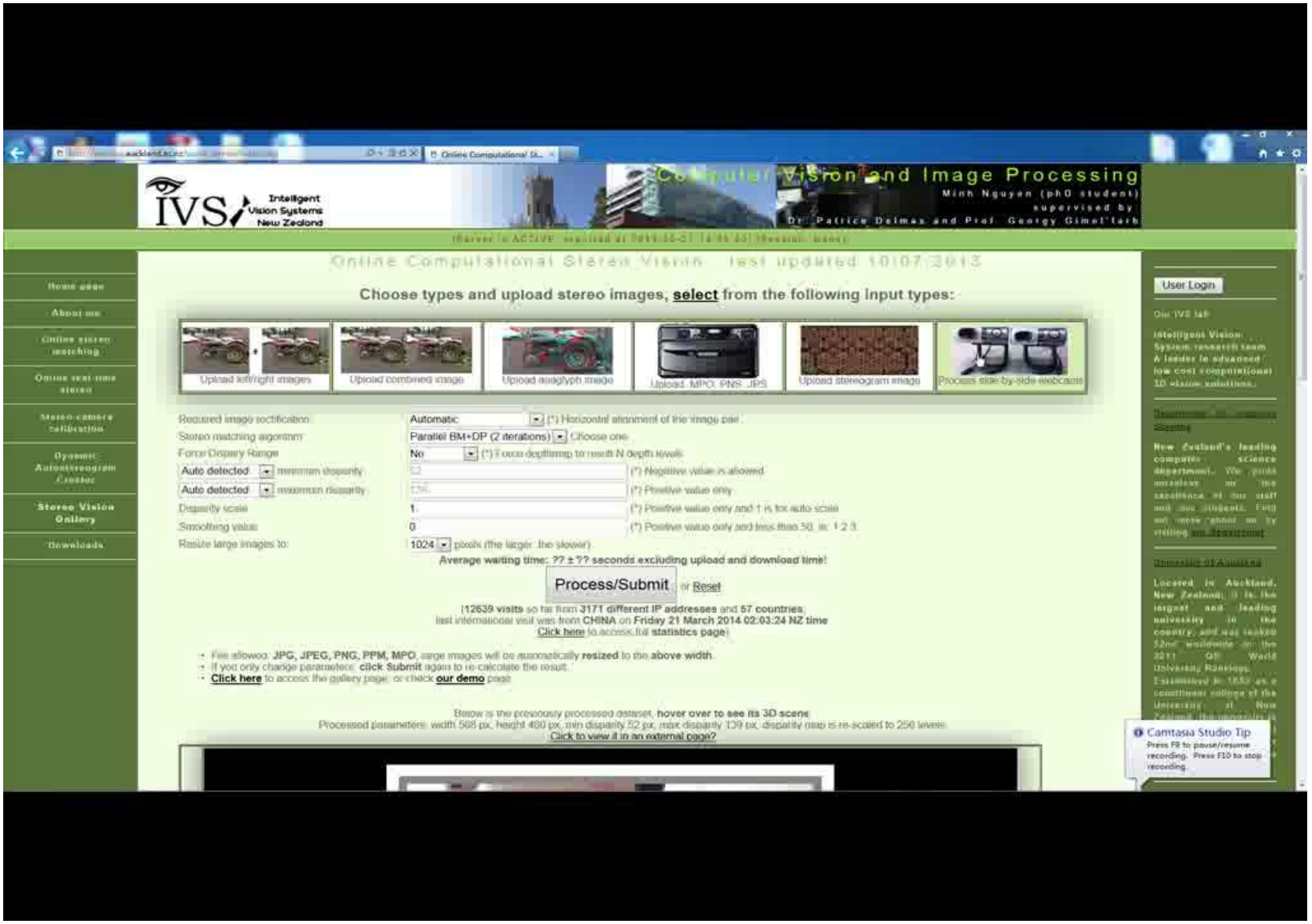
Welcome to the Middlebury Stereo Vision Page, formerly located at www.middlebury.edu/stereo. This website accompanies our taxonomy and comparison of two-frame stereo correspondence algorithms [1]. It contains:

- An on-line evaluation of current algorithms
- Many stereo datasets with ground-truth disparities
- Our stereo correspondence software
- An on-line submission script that allows you to evaluate your stereo algorithm in our framework

How to cite the materials on this website:
We grant permission to use and publish all images and numerical results on this website. If you report performance results, we request that you cite our paper [1]. Instructions on how to cite our datasets are listed on the datasets page. If you want to cite this website, please use the URL "**vision.middlebury.edu/stereo/**".

References:

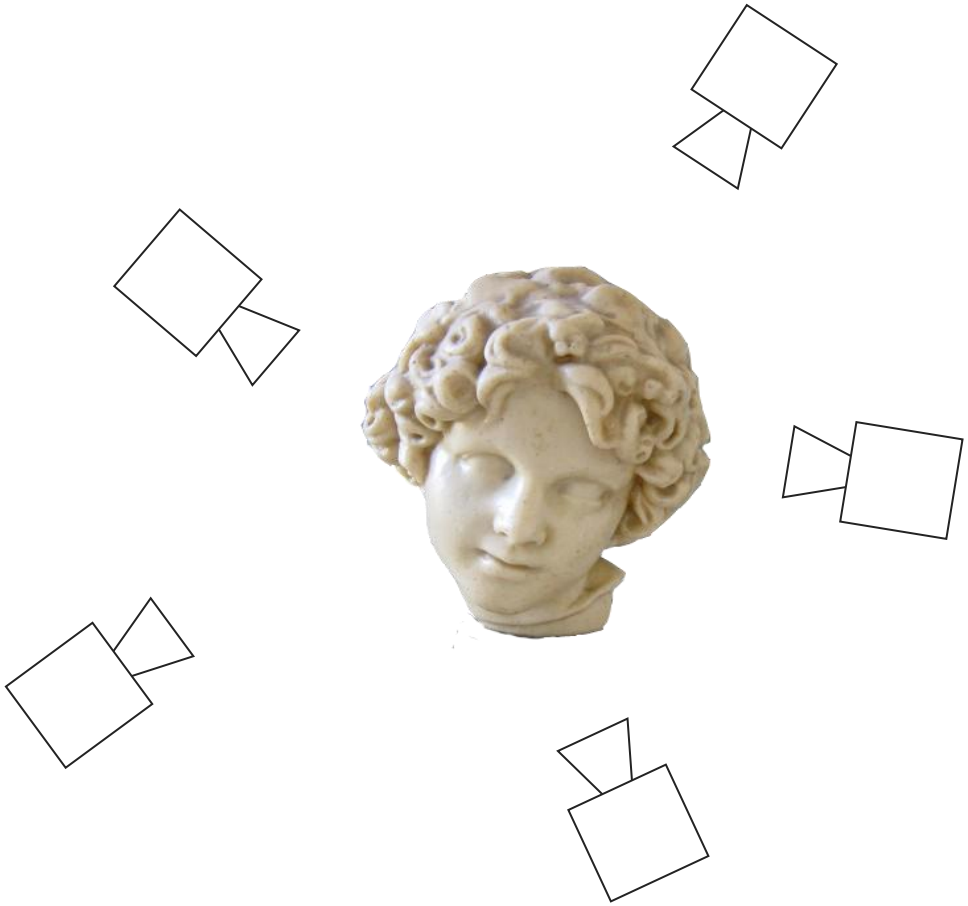
[1] D. Scharstein and R. Szeliski. A taxonomy and evaluation of dense two-frame stereo correspondence algorithms. *International Journal of Computer Vision*, 47(1/2/3):7-42, April-June 2002.
Microsoft Research Technical Report MSR-TR-2001-81, November 2001.

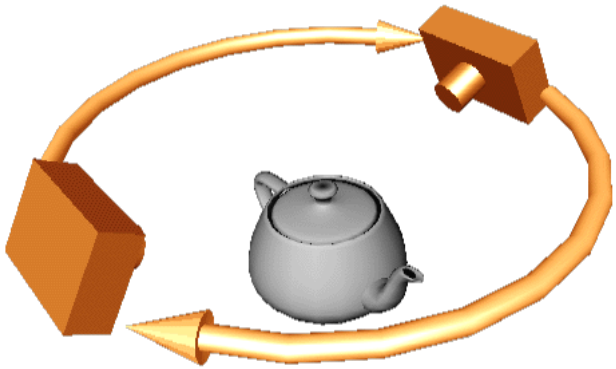


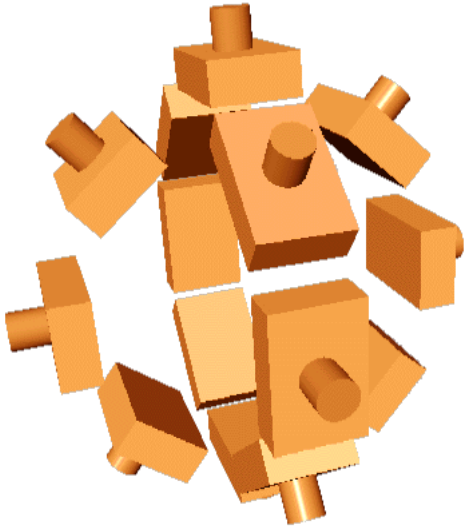
https://www.ivs.auckland.ac.nz/quick_stereo/

3D Construction

Multiple cameras
Sequence Images













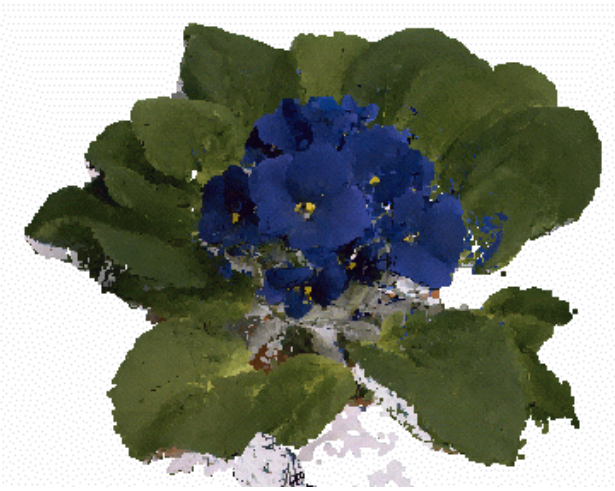


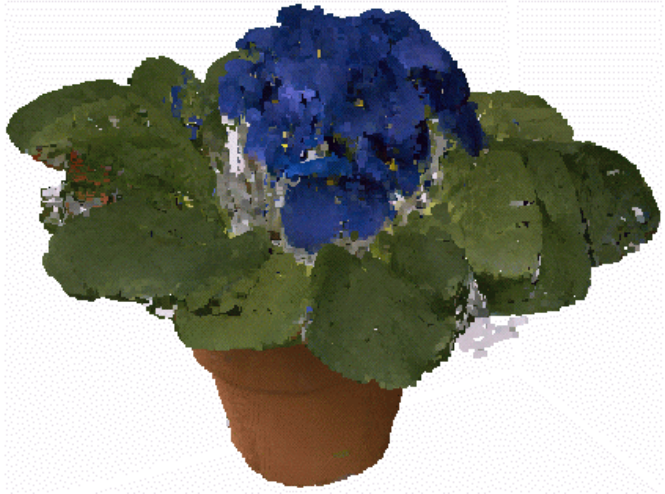


Construction Results









Construction Results

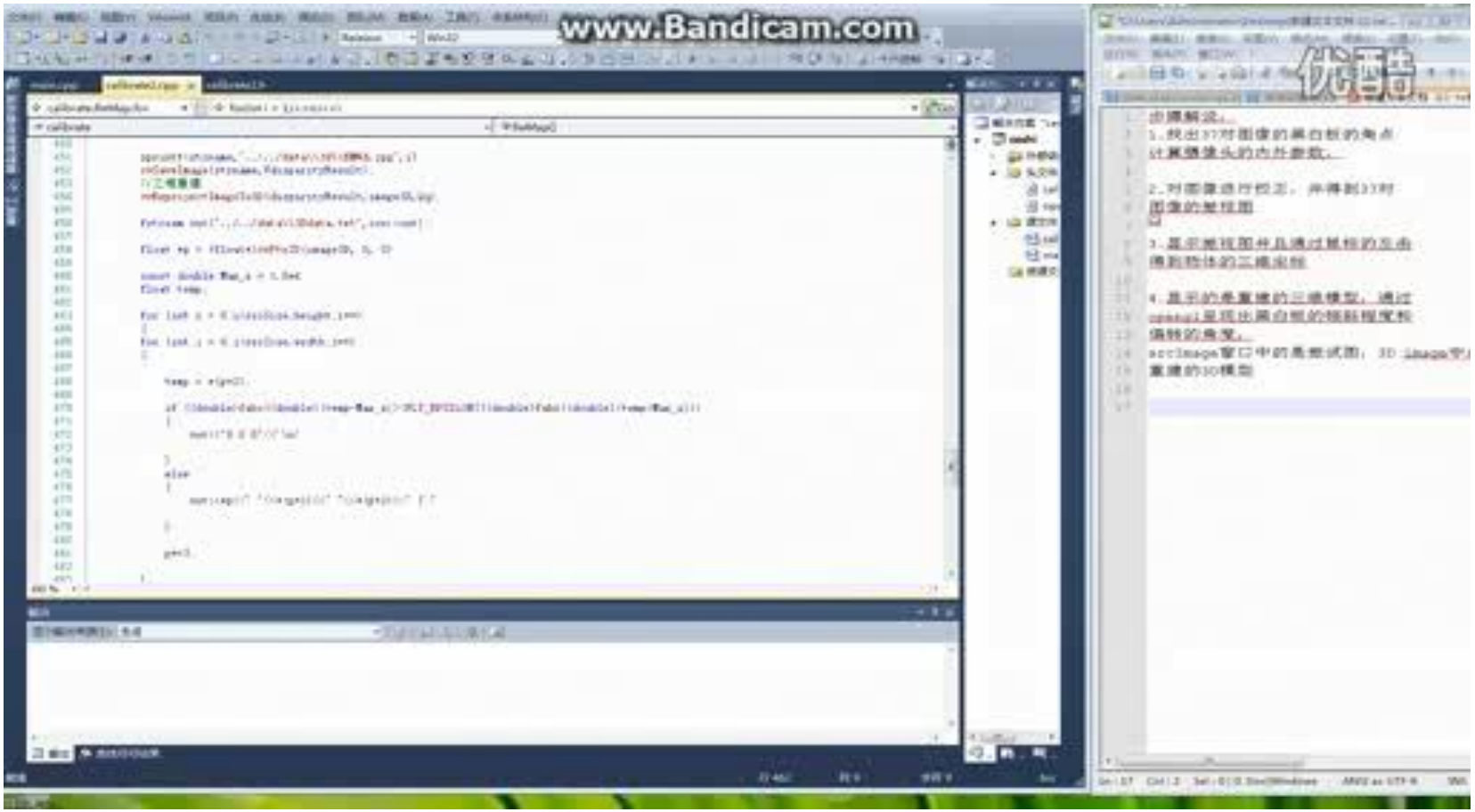






Demo

process



Demo

3D scene modeling



Incremental Reconstruction of Manifold Surface from Sparse Visual Mapping

Demo

Building Rome on a cloudless day

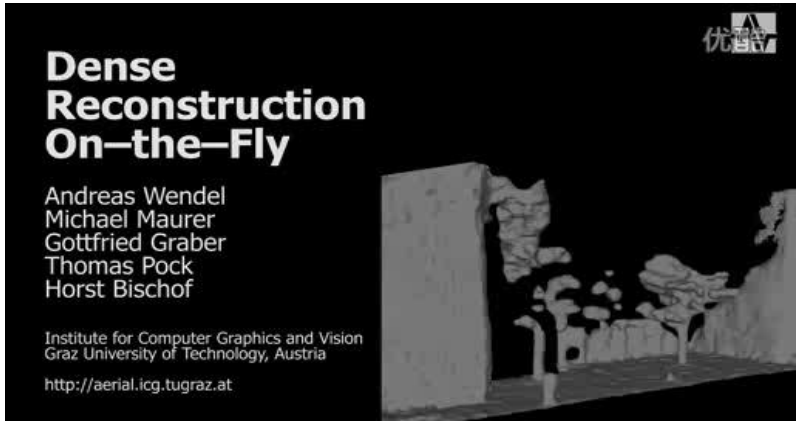


Fig. 1. Example models of our method from Rome (left) and Berlin (right) computed in less than 24 hours from automatically registered subsets of photo collections of 2.9 million and 2.8 million images respectively

https://tv.sohu.com/v/cGwvNjQwMzIzNS8yMDE1NzAyOS5zaHRtbA==.html

Demo

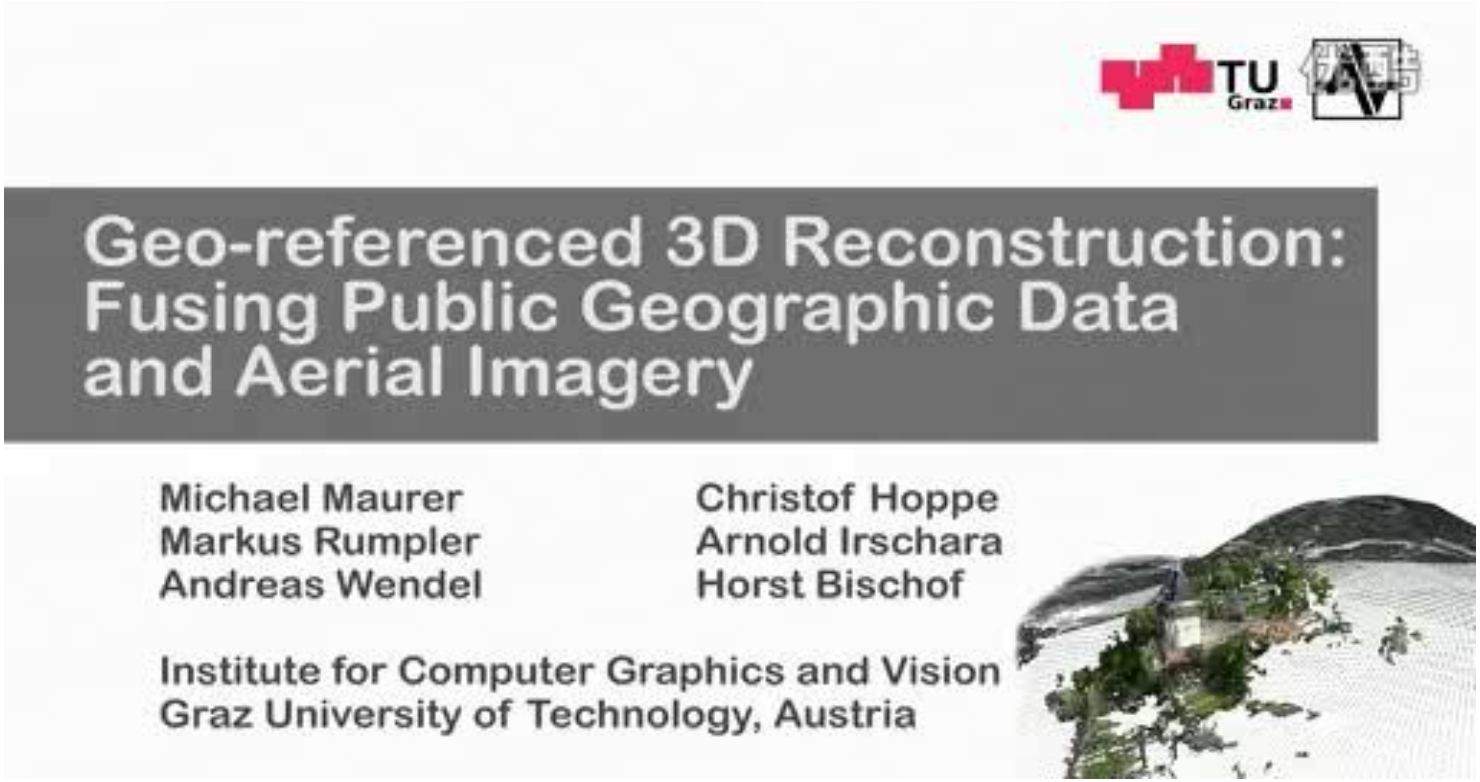
Dense reconstruction on-the-fly



https://tv.sohu.com/v/cGwvNjQwMzIzNS8xNjk4MTUzMy5zaHRtbA==.html

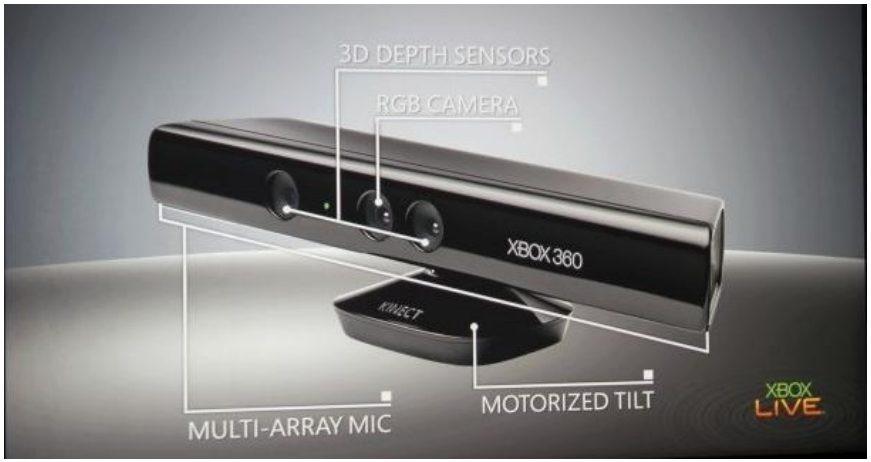
Demo

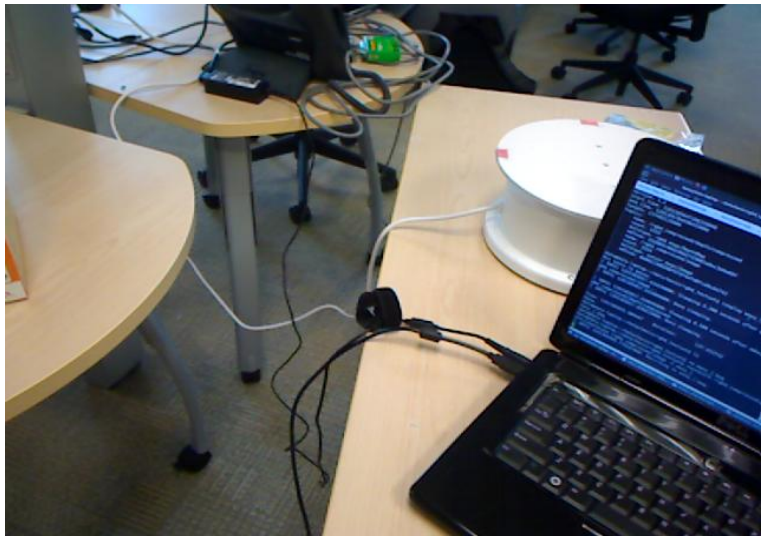
Geo-referenced 3D reconstruction: fusing public geographic data and aerial imagery



扩充内容

- kinect

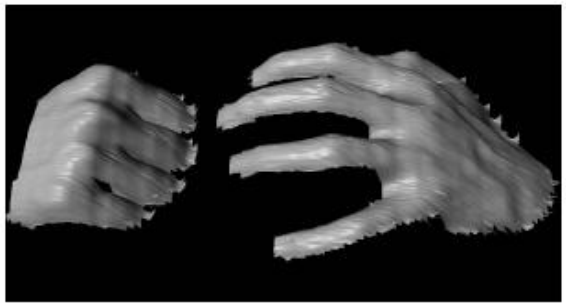

3D DEPTH SENSORS
RGB CAMERA
XBOX 360
KINECT
MULTI-ARRAY MIC
MOTORIZED TILT
XBOX LIVE



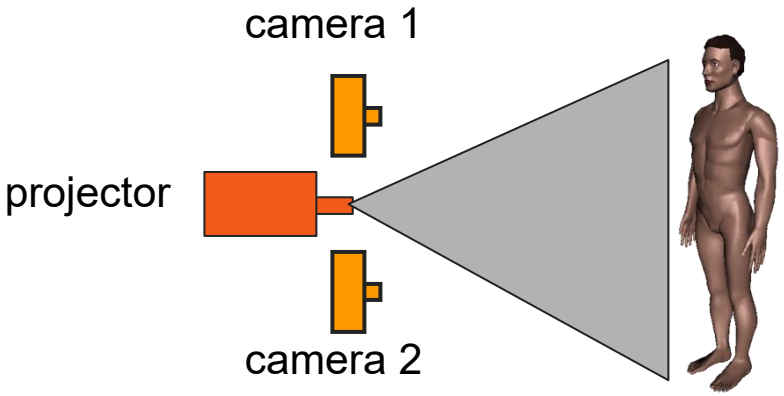


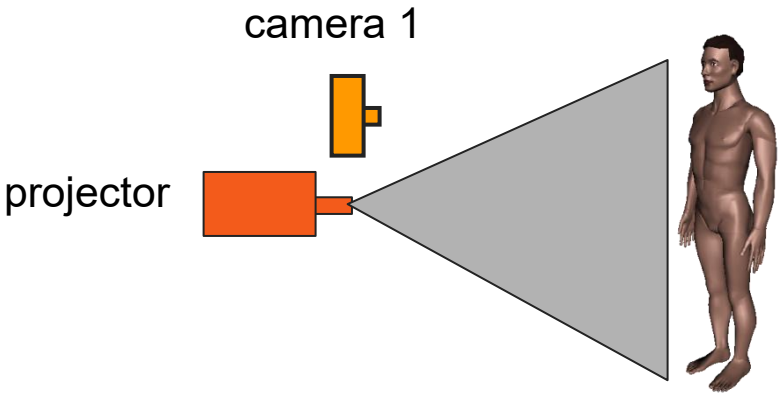
Active stereo with structured light





Li Zhang's one-shot stereo





Li Zhang, Brian Curless, and Steven M. Seitz. Rapid Shape Acquisition Using Color Structured Light and Multi-pass Dynamic Programming. In *Proceedings of the 1st International Symposium on 3D Data Processing, Visualization, and Transmission (3DPVT)*, Padova, Italy, June 19-21, 2002, pp. 24-36.

Active stereo with structured light

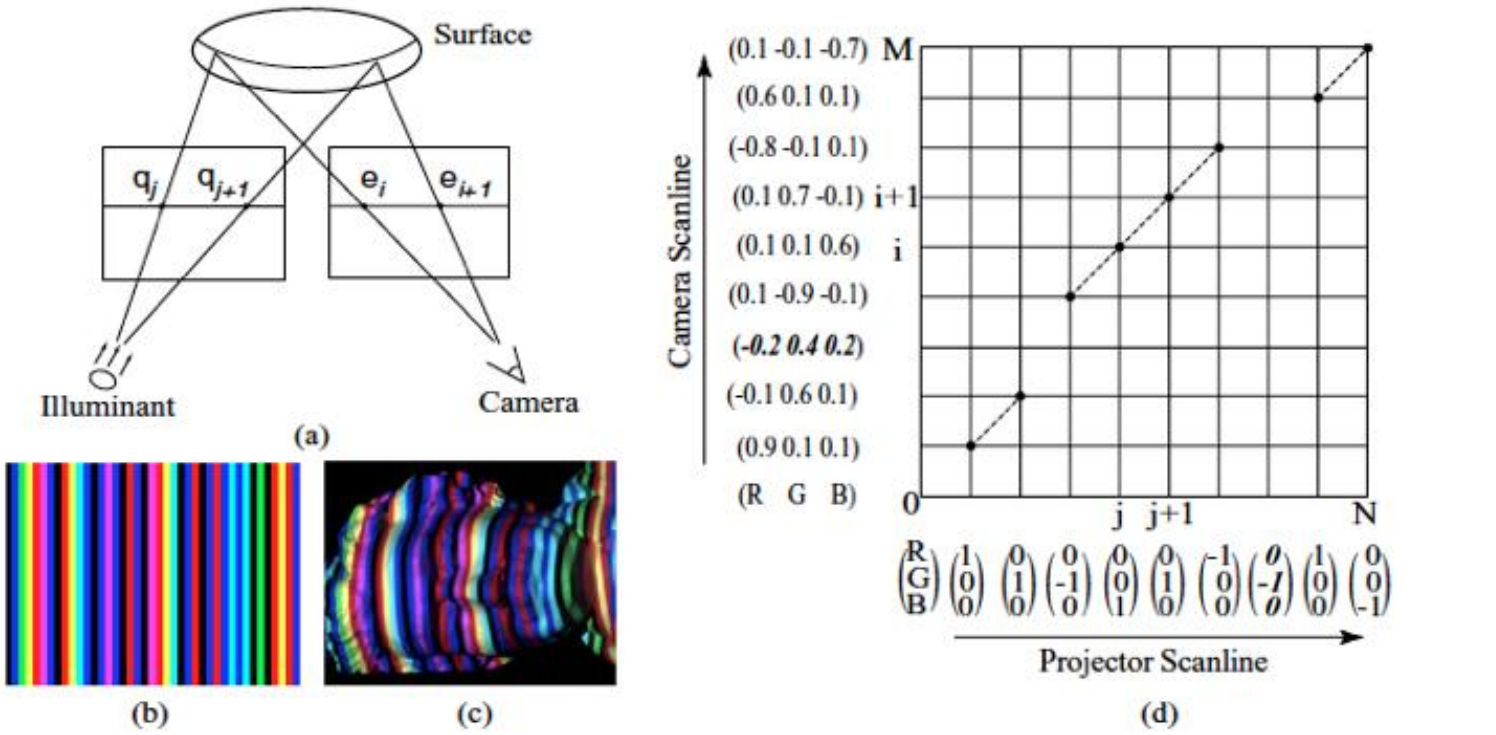
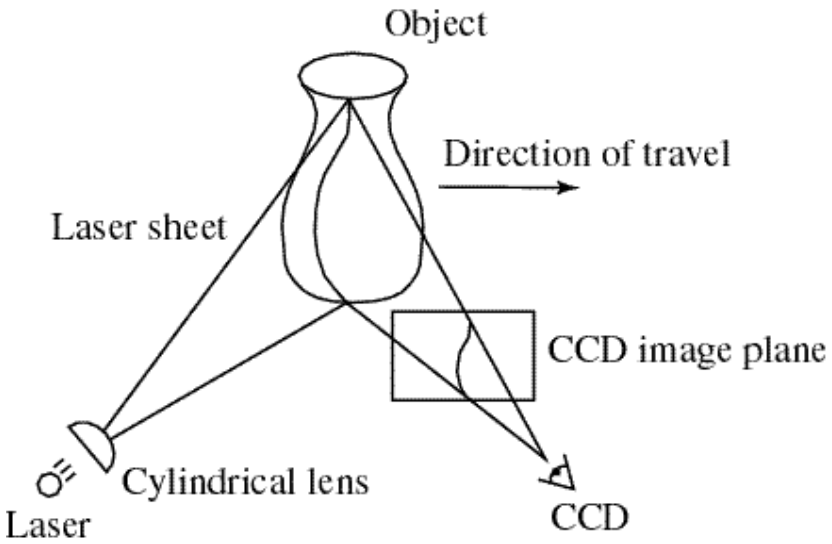
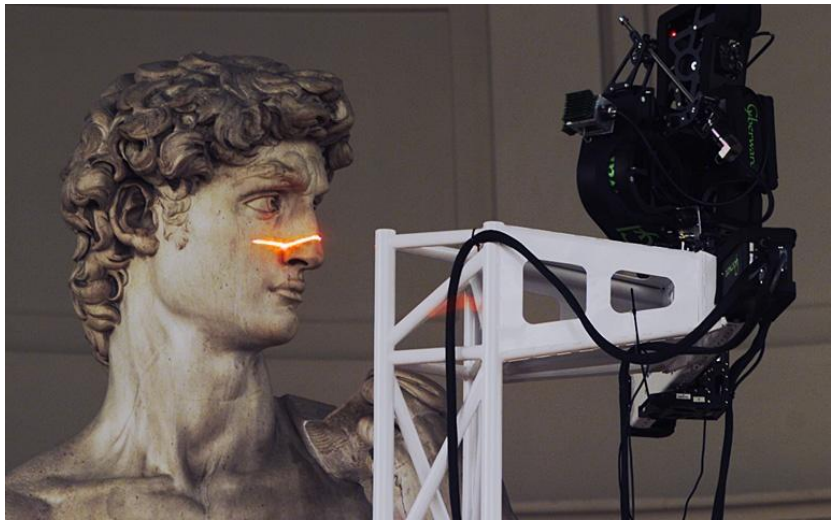


Figure 2. Summary of the one-shot method. (a) In optical triangulation, an illumination pattern is projected onto an object and the reflected light is captured by a camera. The 3D point is reconstructed from the relative displacement of a point in the pattern and image. If the image planes are rectified as shown, the displacement is purely horizontal (one-dimensional). (b) An example of the projected stripe pattern and (c) an image captured by the camera. (d) The grid used for multi-hypothesis code matching. The horizontal axis represents the projected color transition sequence and the vertical axis represents the detected edge sequence, both taken for one projector and rectified camera scanline pair. A match represents a path from left to right in the grid. Each vertex (j, i) has a score, measuring the consistency of the correspondence between e_i, the color gradient vectors shown by the vertical axis, and q_j, the color transition vectors shown below the horizontal axis. The score for the entire match is the summation of scores along its path. We use dynamic programming to find the optimal path. In the illustration, the camera edge in bold italics corresponds to a false detection, and the projector edge in bold italics is missed due to, e.g., occlusion.

Li Zhang, Brian Curless, and Steven M. Seitz

Laser scanning



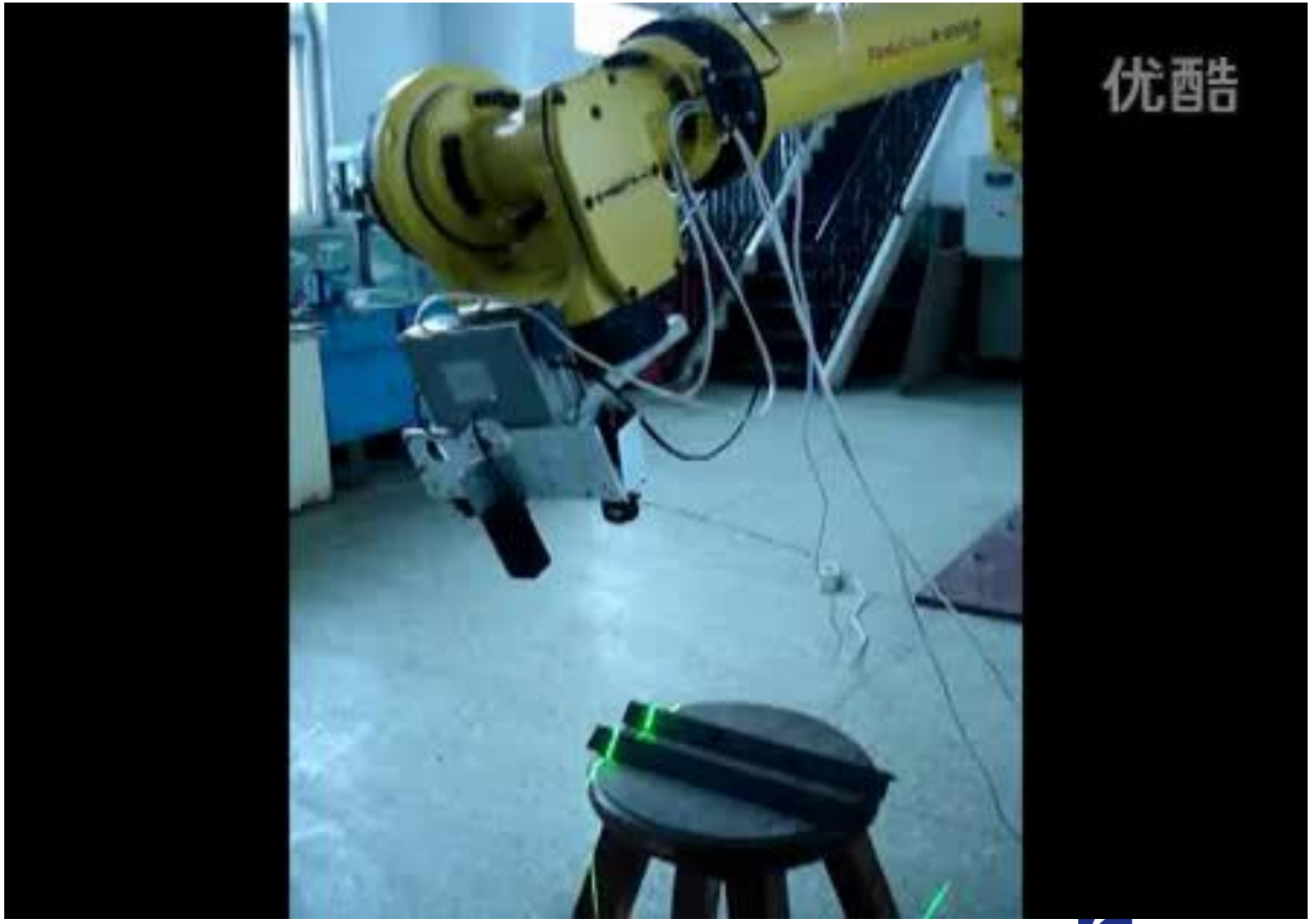


Digital Michelangelo Project
http://graphics.stanford.edu/projects/mich/

- Optical triangulation
  - Project a single stripe of laser light
  - Scan it across the surface of the object
  - This is a very precise version of structured light scanning

Source: S. Seitz

Active stereo with structured light



- Learning based method



The end !